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Structure and evolution of Alanine/Serine Decarboxylases and the engineering of theanine production

Hao Wang, Biying Zhu, Siming Qiao, Chunxia Dong, Xiaochun Wan, Weimin Gong, Zhaoliang Zhang

Department of Life Sciences and Medicine, University of Science and Technology of China, Hefei, China • State Key Laboratory of Tea Plant Biology and Utilization, Anhui Agricultural University, Hefei, Anhui 230036, China

Abstract

Ethylamine (EA), the precursor of theanine biosynthesis, is synthesized from alanine decarboxylation by alanine decarboxylase (AlaDC) in tea plants. AlaDC evolves from serine decarboxylase (SerDC) through neofunctionalization and has lower catalytic activity. However, lacking structure information hinders the understanding of the evolution of substrate specificity and catalytic activity. In this study, we solved the X-ray crystal structures of AlaDC from *Camellia sinensis* (CsAlaDC) and SerDC from *Arabidopsis thaliana* (AtSerDC). Tyr^{341} of AtSerDC or the corresponding Tyr^{336} of CsAlaDC is essential for their enzymatic activity. Tyr^{111} of AtSerDC and the corresponding Phe^{106} of CsAlaDC determine their substrate specificity. Both CsAlaDC and AtSerDC have a distinctive zinc finger and have not been identified in any other Group II PLP-dependent amino acid decarboxylases. Based on the structural comparisons, we conducted mutation screen of CsAlaDC. The results indicated that the mutation of L110F or P114A in the CsAlaDC dimerization interface significantly improved the catalytic activity by 110% and 59%, respectively. Combining a double mutant of $CsAlaDC^{L110F/P114A}$ with theanine synthetase increased theanine production 672% in an *in vitro* system. This study provides the structural basis for the substrate selectivity and catalytic activity of CsAlaDC and AtSerDC and provides a route to more efficient biosynthesis of theanine.

eLife assessment

This study reports a comparative biochemical and structural analysis of two PLP decarboxylase enzymes from plants. The work is **useful** because of the potential application of these enzymes in industrial theanine production. The structure provides a **solid** basis for understanding substrate specificity but some aspects of the work are **incomplete**. The paper will be of interest to enzymologists studying PLP enzymes and those working on enzyme engineering in plants.

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Introduction

Tea is one of the most popular natural non-alcoholic beverages consumed worldwide. About 2 billion cups of tea are consumed worldwide daily (1). Theanine (γ-glutamylethylamide) is a unique non-protein amino acid and the most abundant free amino acid in tea plants (*Camellia sinensis*). It accounts for more than 50% of the total free amino acids and approximately 1-2% of the dry weight of the new shoots of tea plants (2). Theanine is the secondary metabolite conferring the umami taste of tea infusion and also balances the astringency and bitterness of tea infusion caused by catechins and caffeine (3). It has also many health-promoting functions, including neuroprotective effects, enhancement of immune functions, and potential anti-obesity capabilities, among others (2, 4–9). Therefore, theanine content is highly correlated with green tea quality (10).

Theanine is synthesized from ethylamine (EA) and glutamate (Glu) by theanine synthetase (11, 12). Importantly, the large amount of theanine biosynthesis is determined by the high availability of EA in tea plants (13). Therefore, the evolution of EA biosynthesis in tea plants provided the basis for theanine biosynthesis and the formation of tea quality. EA is synthesized from alanine decarboxylation by alanine decarboxylase (CsAlaDC) (14) (**Figure 1A**). Indeed, CsAlaDC expression level and catalytic activity largely determine the theanine accumulation level in tea plants (15). As a novel gene, *CsAlaDC* had not been reported in other plant species before it was identified in tea plants. Previous studies indicated that *AlaDC* originated from the serine decarboxylase gene (*SerDC*) by gene duplication in tea plants (16). However, despite sharing highly conserved amino acid sequences, CsAlaDC and CsSerDC specifically catalyze the decarboxylation of alanine and serine, respectively (16) (**Figure 1A and B**). In addition, CsAlaDC exhibited a much lower enzymatic activity compared with CsSerDC (16). However, the structural basis and key amino acids underlying the evolution of the substrate specificity and enzymatic activity of CsAlaDC are unknown.

SerDC belongs to the Group II pyridoxal-5-phosphate (PLP)-dependent decarboxylase superfamily (17). These Group II amino acid decarboxylases produce many bioactive secondary metabolites and signaling molecules (18–20). In plants, SerDC catalyzes the biosynthesis of ethanolamine from serine, which was first functionally characterized in *Arabidopsis thaliana* (21) (**Figure 1A**). Ethanolamine is an important metabolite in the synthesis of phosphatidylethanolamine and phosphatidylcholine, two main phospholipids involved in maintaining eukaryotic membrane structures in eukaryotic cell membranes (22–24). Furthermore, the study has shown that SerDC is essential in the embryogenesis of *Arabidopsis* (25). SerDC is of vital importance for plant growth and development, but the mechanism of substrate recognition and catalytic activity of SerDC also remains unclear.

In this work, we attempt to understand the mechanism of functional divergence of plant AlaDC and SerDC by analyzing their structural characteristics. Here, we obtained the X-ray crystal structure of CsAlaDC and AtSerDC. According to the crystal structure, we found a distinctive zinc finger structure located at both CsAlaDC and AtSerDC, which has not been identified in any other Group II PLP-dependent amino acid decarboxylases that have been previously characterized. By comparing the substrate binding pockets, we identified Phe^{106} of CsAlaDC and Tyr^{111} of AtSerDC as the crucial sites for substrate specificity. By conducting mutation screening based on the protein structures, we identified the amino acids repressing the catalytic activity and discovered that CsAlaDC$^{L110F/P114A}$ exhibited a 2.3-fold increase in catalytic activity compared to that of CsAlaDC and enhanced the engineering of theanine production *in vitro*.

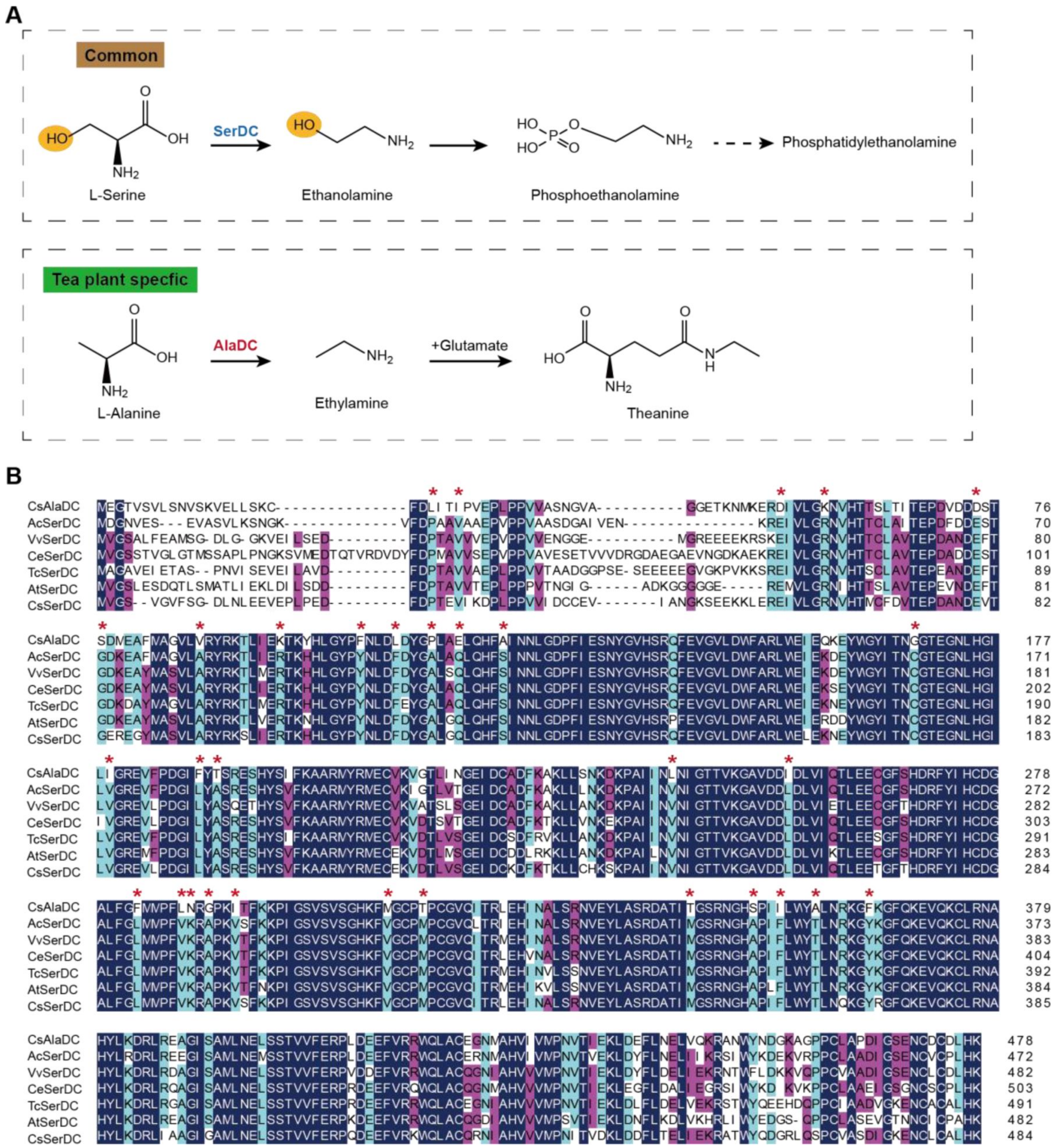

Figure 1.

Metabolic pathways and sequence analysis of CsAlaDC and SerDCs in plants.

(**A**) The decarboxylation of serine and alanine in plants. SerDC, serine decarboxylase; AlaDC, alanine decarboxylase. (**B**) Multiple alignment of the amino acid sequences of AlaDC and SerDCs. The amino acid sequences of the 6 SerDCs from kiwifruit (*Actinidia chinensis*), grape (*Vitis vinifera*), coffee (*Coffea eugenioides*), cocoa (*Theobroma cacao*) and Arabidopsis. Primary (100%), secondary (80%), and tertiary (60%) conserved percent of similar amino acid residues were shaded in deep blue, light blue and cheer red, respectively. "*" indicated amino acid residues mutated only in CsAlaDC.

Results

Enzymatic properties of CsAlaDC, AtSerDC, and CsSerDC

CsAlaDC originates from CsSerDC by gene duplication and neofunctionalization in tea plants, but they catalyze different metabolic processes (**Figure 1A**). We performed a multiple sequence alignment of the amino acid sequences of the 6 SerDCs from kiwifruit (*Actinidia chinensis*), grape (*Vitis vinifera*), coffee (*Coffea eugenioides*), cocoa (*Theobroma cacao*), Arabidopsis and tea plants and CsAlaDC (**Figure 1B**). The results showed that the amino acid sequences of CsAlaDC and SerDCs were highly conserved, but CsAlaDC has amino acid mutations in highly conserved regions compared with SerDCs.

Next, to verify the substrate specificity and enzyme activity, we conducted enzyme activity detection and enzyme kinetics analysis for CsAlaDC, CsSerDC and AtSerDC. The 5' truncated *CsAlaDC* and *AtSerDC* were inserted into the pET22b and pET28a expression vectors to generate recombinant plasmids pET22b-*CsAlaDC* and pET28a-*AtSerDC*, correspondingly. Additionally, the full-length protein of CsSerDC was inserted into the pET28a expression vector to generate the recombinant plasmid pET28a-*CsSerDC*. Subsequently, the recombinant proteins were expressed in *Escherichia coli* and purified via nickel affinity chromatography. The purified proteins were examined through SDS-PAGE analysis (**Figure 2A**), supplemented by size-exclusion chromatography assessments (Supplementary figure 1A), it seems that the protein manifests in an oligomeric configuration within the solution.

To verify the substrate specificity of CsAlaDC, AtSerDC, and CsSerDC, enzyme activity assays were conducted using Ala and Ser as substrates, followed by product identification via UPLC analysis. The results showed that CsAlaDC can effectively catalyze the decarboxylation of alanine to generate EA, whereas its catalytic efficacy towards serine was relatively inferior. Conversely, AtSerDC and CsSerDC selectively catalyzed serine decarboxylation to yield ethanolamine but did not exhibit activity towards alanine (**Figure 2B**). These findings confirmed that CsAlaDC and SerDCs do not have promiscuous decarboxylase activity on Ala and Ser.

The kinetic properties of CsAlaDC, AtSerDC, and CsSerDC were determined through the use of corresponding substrates (**Figure 2C**), and the results are presented in **Table 1**. The kinetic parameter *Km* of CsAlaDC was determined to be 1.215 mM, which is similar to that of AtSerDC and CsSerDC at 1.522 mM and 2.364 mM, respectively. However, AtSerDC has a *Vmax* of 7.053 μmo L^{-1} s^{-1}, which is 4-fold that of CsAlaDC's *Vmax* of 1.709 μmo L^{-1} s^{-1}; while CsSerDC has a *Vmax* of 9.031 μmo L^{-1} s^{-1}, which is 5.3-fold that of CsAlaDC. The findings suggest that the enzymatic catalytic efficiency of CsSerDC and AtSerDC is approximately triple compared to that of CsAlaDC, and the *Vmax* of CsAlaDC is considerably lower than that of both CsSerDC and AtSerDC. The observed disparity in *Vmax* suggests that CsAlaDC and SerDCs may exhibit discrepancies in substrate-to-product conversion rates under saturated substrate conditions. It is plausible that SerDCs possess a higher turnover rate, facilitating the rapid conversion of substrate to product and subsequent release of the enzyme. Another possibility is that SerDCs demonstrate enhanced stability, thereby maintaining superior catalytic activity even at higher substrate concentrations, consequently leading to a higher *Vmax* value. Due to the similarity in catalytic activity between CsSerDC and AtSerDC, we chose the more representative AtSerDC for further analysis.

The overall structures of CsAlaDC and AtSerDC

To enhance our comprehension of CsAlaDC and AtSerDC, we conducted structural analyses of these two proteins. Through optimization of crystallization conditions, we successfully determined the crystal structures of CsAlaDC, CsAlaDC-EA, and AtSerDC at resolutions of 2.50 Å, 2.60 Å and 2.85

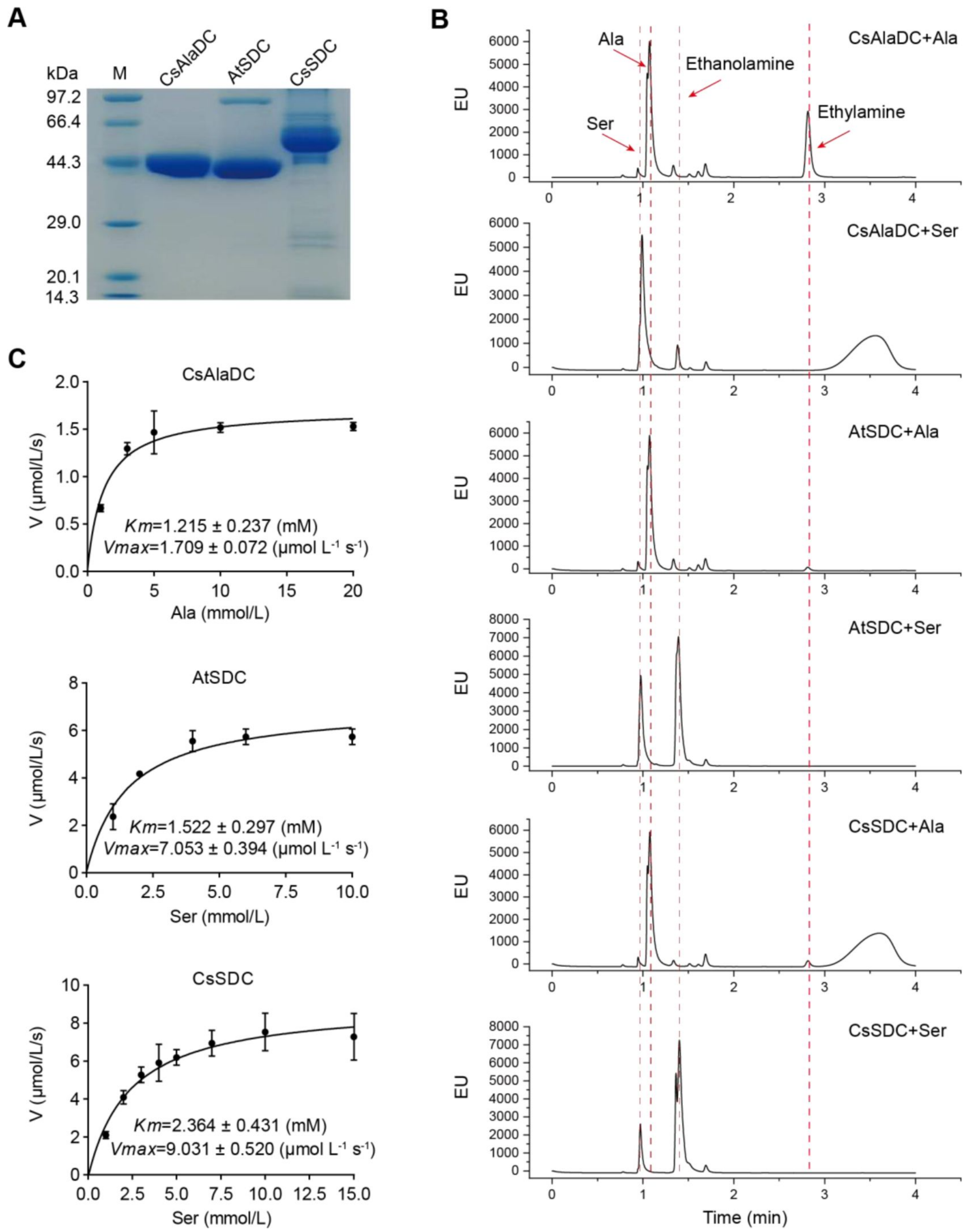

Figure 2.

Purification and characterization of CsAlaDC, AtSerDC, and CsSerDC.

(**A**) Identification of CsAlaDC, AtSerDC, and CsSerDC using SDS-PAGE. (**B**) Detection of enzyme activities of CsAlaDC, AtSerDC, and CsSerDC by UPLC. (**C**) Reaction rates of substrates with different concentrations catalyzed by CsAlaDC, AtSerDC, and CsSerDC.

Enzyme	Substrate	*Km* (mM)	*Vmax* (μmol L^{-1} s^{-1})	*Kcat* (s^{-1})	*Kcat/Km* (s^{-1} M^{-1})
CsAlaDC	Alanine	1.215 ± 0.237	1.709 ± 0.072	0.068	55.7
AtSerDC	Serine	1.522 ± 0.297	7.053 ± 0.297	0.281	184.6
CsSerDC	Serine	2.364 ± 0.431	9.031 ± 0.520	0.361	152.7

Table 1.

Kinetic parameters of CsAlaDC, AtSerDC and CsSerDC

Å, respectively (Supplementary table 1).

The overall structure of CsAlaDC or AtSerDC is homodimeric with two subunits exhibiting an asymmetrical arrangement (**Figure 3A** and **3D**). The monomer of CsAlaDC or AtSerDC is divided into three distinct structural domains: an N-terminal domain (N-terminal –104 aa in CsAlaDC, N-terminal –109 aa in AtSerDC), a large domain (105-364 aa in CsAlaDC, 110-369 aa in AtSerDC), and a small C-terminal domain (365 aa-C-terminus in CsAlaDC, 370-C-terminus as in AtSerDC), colored light pink, khaki, and sky blue, respectively. Compared to the full-length protein, the N-terminal structures of CsAlaDC and AtSerDC were truncated by 60 and 65 amino acid residues, respectively. The N-terminal domain of the truncated protein contains a long α-helix, which is connected to the large domain through a long loop. The helix of one subunit is antiparallel to the corresponding helix in the neighboring subunit, forming a clamp. Therefore, the N-terminal domain may not be an independent folding unit and is likely only stable in dimers. The large domain contains a seven-stranded mixed β-sheet surrounded by eight α-helices, while the C-terminal domain is composed of a three-stranded antiparallel β-sheet and three α-helices (Supplementary figure 1). The catalytic site of the enzyme is positioned within a superficial crevice at the junction of two subunits forming a dimeric arrangement. The amino acid residues derived from both subunits participate in the binding of the PLP, effectively anchoring it to the active site. In the catalytic site, one monomer accommodates PLP, while the other monomer also contributes to PLP binding and enzymatic function (**Figure 3A** and **3D**).

Notably, our investigation has revealed the presence of a distinctive zinc finger structure (as depicted in **Figure 3A** and **3D**) located at the C-terminus of CsAlaDC and AtSerDC. This structure is composed of a loop structure spanning 17 amino acid residues, wherein coordination of Zn^{2+} is facilitated by three Cys residues and one His residue. Importantly, this particular configuration is exclusive to the two proteins under examination and has not been identified in any other Group II PLP-dependent amino acid decarboxylases that have been previously characterized (26, 27).

The crystal structures of CsAlaDC-EA complex and AtSerDC were obtained and further analyzed. In the former complex, PLP bound to Lys^{309} via a Schiff base linkage (internal aldimine form), and the pyridine moiety of PLP is positioned between the imidazole ring of His^{196} and the methyl group of Ala^{279} in a parallel orientation to the imidazole ring. The carboxylic group of Asp^{277} stabilizes the N1 of PLP by a salt bridge interaction, providing the latter with the strong electrophilicity necessary to stabilize carbon ion intermediates during enzymatic catalysis. The O3 forms hydrogen bonds with Thr^{247} and Lys^{309}. Additionally, His^{308}, Gly^{169}, Thr^{170}, Lys^{309}, and Ser^{347}*(“*” denotes the amino acids on adjacent subunits) establish a hydrogen bonding network with the phosphate group of PLP, consequently fortifying its attachment to the protein (**Figure 3C**). Similarly, in the latter structure, the pyridine ring of PLP was sandwiched between the imidazole ring of His^{201} and the methyl group of Ala^{284}. The N1 of PLP formed a salt bridge with Asp^{282}, while the O3 formed hydrogen bonds with Thr^{252}. The phosphate group of PLP established hydrogen bonding interactions with Lys^{314}, His^{313}, Gly^{174}, Thr^{175}, and Ser^{352}* (**Figure 3E**). The aforementioned amino acid residues are observed across various Group II PLP-dependent amino acid decarboxylases and exhibit considerable stability (see further discussion below), suggesting that they could share common features in terms of catalytic mechanisms.

Factors affecting CsAlaDC and AtSerDC activity

The Group II PLP-dependent amino acid decarboxylases are dimeric enzymes with each monomer containing an active site located within a shallow cavity (28). A flexible loop of amino acids originating from one monomer extends into the active site of the other monomer and plays a crucial role in catalysis. The catalytic loop in AtSerDC, consisting of amino acid residues 328-341, is well-structured and exhibits clear electron density. However, the corresponding loop in CsAlaDC is disordered (**Figure 3F**). The aforementioned loop harbors a conserved Tyr residue that is believed to function as a proton donor of the carbanion in the catalytic process (29). In

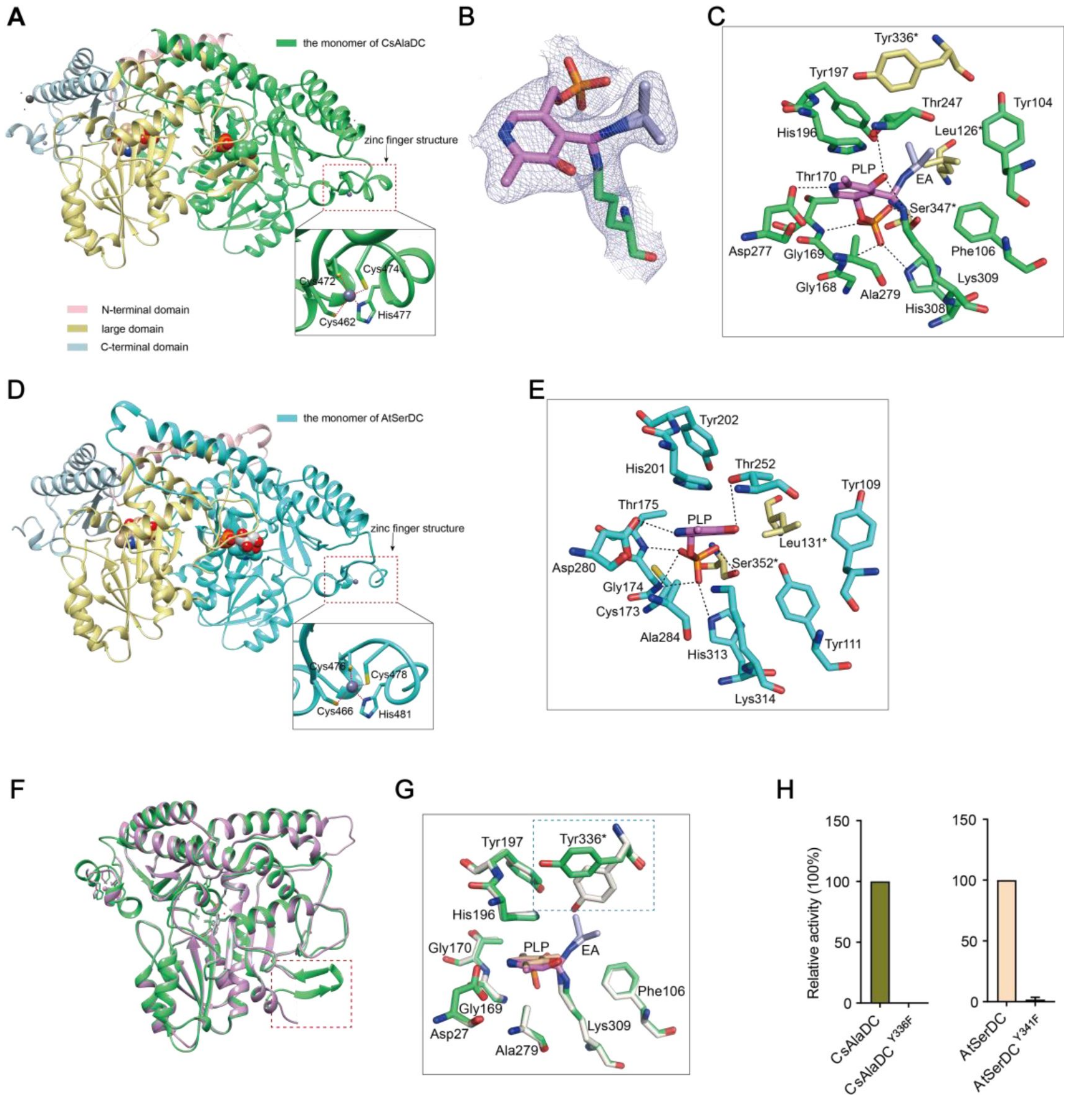

Figure 3.

Crystal structures of CsAlaDC and AtSerDC.

(**A**) Dimer structure of CsAlaDC. The color display of the N-terminal domain, large domain, and C-terminal domains of chain A is shown in light pink, khaki and sky blue, respectively. Chain B is shown in spring green. The PLP molecule is shown as a sphere model. The zinc finger structure at the C-terminus of CsAlaDC is indicated by the red box. The gray spheres represent zinc ions, while the red dotted line depicts the coordination bonds formed by zinc ions with cysteine and histidine. (**B**) The 2Fo-Fc electron density maps of K309-PLP-EA (contoured at 1σ level). The PLP is shown in violet, the K309 is shown in spring green, and the EA is shown in lightblue. (**C**) Active center of the CsAlaDC-EA complex, with hydrogen bonds denoted by black dotted lines. "*" denotes the amino acids on adjacent subunits. (**D**) Dimer structure of AtSerDC. The color display of the N-terminal domain, large domain, and C-terminal domains of chain A is shown in light pink, khaki and sky blue, respectively. Chain B is shown in cyan. The PLP molecule is shown as a sphere model. The zinc finger structure at the C-terminus of AtSerDC is indicated by the red box. The gray spheres represent zinc ions, while the red dotted line depicts the coordination bonds formed by zinc ions with cysteine and histidine. (**E**) Active center of the AtSerDC, with hydrogen bonds denoted by black dotted lines. "*" denotes the amino acids on adjacent subunits. (**F**) The monomers of CsAlaDC and AtSerDC are superimposed. CsAlaDC is depicted in spring green, while AtSerDC is shown in plum. The conserved amino acid catalytic loop is indicated by the red box. (**G**) Amino acid residues of the active center in CsAlaDC apo and CsAlaDC-EA complex are superimposed. CsAlaDC apo is shown in floral white, while CsAlaDC-EA complex is shown in spring green. (**H**) The relative activity of wild-type CsAlaDC and its Y336F mutant (left), as well as wild-type AtSerDC and its Y341F mutant (right) is shown.

particular, CsAlaDC Tyr^{336} and AtSerDC Tyr^{341} correspond to that Tyr residue (Supplementary figure 2). Upon superimposing the active site residues of CsAlaDC apo and CsAlaDC-EA complex, it was observed that the hydroxyl group of Tyr^{336*} in the CsAlaDC-EA complex exhibits a substantial 60-degree deviation along the Cα-Cβ bond (**Figure 3G**). To assess the function of the Tyr, we substituted the corresponding Tyr residues in CsAlaDC and AtSerDC with Phe. The resulting CsAlaDC^{Y336F} and AtSerDC^{Y341F} mutants were then exposed to Ala and Ser, respectively. We measured the production of EA or ethanolamine in the reaction mixtures. Our findings indicate that these mutants catalyze abortive decarboxylation, as evidenced by the absence of detectable EA and only a small amount of ethanolamine observed in the reaction mixture (**Figure 3H**). This result suggested this Tyr is required for the catalytic activity of CsAlaDC and AtSerDC.

Identification of key amino acids for the substrate specificity

The superposition of amino acid residues in the substrate binding pocket of CsAlaDC and AtSerDC revealed that all residues are identical, except for the substitution of Tyr at position 111 in AtSerDC with Phe at position 106 in CsAlaDC (**Figure 4A**). This observation suggests a potential role for this specific residue in determining the selective binding of the appropriate substrate.

To gain further insights into the functional role of this residue (Tyr^{111} in AtSerDC or Phe^{106} in CsAlaDC) in selective binding of the appropriate substrate, we identified 563 potential serine decarboxylases in Embryophyta using the amino acid sequences of AtSerDC (Supplementary figure 3A). By comparing the amino acid sequences of these serine decarboxylases, we observed that the corresponding residues to Tyr^{111} or Phe^{106} are located within a conserved motif comprising 9 amino acids (**Figure 4B**). Within this motif, the first two residues, Y (Tyrosine) and P (Proline) are completely conserved across all 563 proteins. However, the third residue where Tyr^{111} in AtSerDC or Phe^{106} in CsAlaDC is positioned exhibits variability among Y, T (Threonine), F (Phenylalanine), V (Valine), A (Alanine), I (Isoleucine), L (Leucine), or other amino acids (**Figure 4B**). Remarkably similar to Tyr^{111}, this residue is predominantly Y in 83.7% of these homologs; conversely, F was found to be present in only 2.3% of these homologs within plant species belonging to Poales, Asterales, Ranunculales, Ericales, Caryophyllales and Nymphaeales orders (**Figure 4C**-**4D**).

To verify the role of Tyr^{111} in AtSerDC or Phe^{106} in CsAlaDC in the selective binding of the appropriate substrate, we introduced mutations in Tyr^{111} to Phe (AtSerDC^{Y111F}) and Phe^{106} to Tyr (CsAlaDC^{F106Y}), followed by *in vitro* enzymatic activity assays. The results revealed that AtSerDC^{Y111F} acquired alanine decarboxylase activity, whereas CsAlaDC^{F106Y} completely lost its alanine decarboxylase activity (**Figure 4E**). Additionally, we generated other mutations of Tyr^{111} in AtSerDC, including AtSerDC^{Y111A}, AtSerDC^{Y111I}, AtSerDC^{Y111L}, AtSerDC^{Y111V} and AtSerDC^{Y111W}. Among these mutants, only AtSerDC^{Y111I} exhibited a marginal level of alanine decarboxylase activity (**Figure 4E**). These findings indicate the essential role of Phe^{106} in the selective binding of alanine for CsAlaDC.

On the other side, both CsAlaDC and CsAlaDC^{F106Y} exhibit a very low level of serine decarboxylase activity (**Figure 4F**). Moreover, compared with AtSerDC, the serine decarboxylase activity of AtSerDC^{Y111F} was about 30% of AtSerDC; AtSerDC^{Y111I} retained lower than 5% of serine decarboxylase activity of AtSerDC; while other mutations, including AtSerDC^{Y111A}, AtSerDC^{Y111L}, AtSerDC^{Y111V} and AtSerDC^{Y111W} abolished the serine decarboxylase activity (**Figure 4F**). These results suggested the Tyr^{111} of AtSerDC is also important for the selective binding of serine for AtSerDC.

To further verify that Phe^{106} of CsAlaDC and Tyr^{111} of AtSerDC were key amino acid residues determining the selective binding of substrates *in planta*, we employed the *Nicotiana benthamiana* transient expression system. To this end, *A. tumefaciens* strain GV3101 (pSoup-p19), carrying recombinant plasmid, was infiltrated into leaves of 5-week-old *N. benthamiana* plants, and the pCAMBIA1305 empty vector was used as the control (EV). Relative mRNA levels were detected

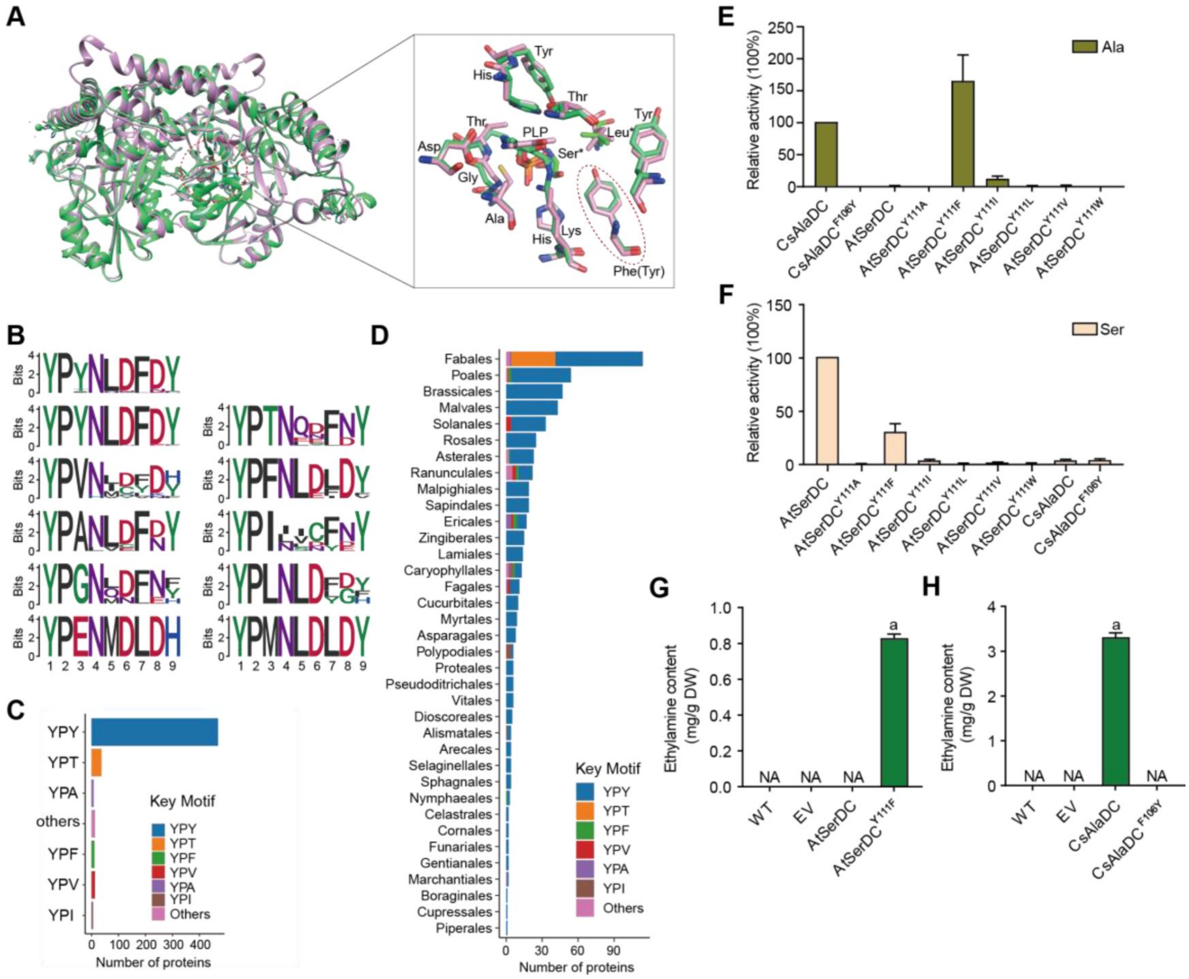

Figure 4.

Key amino acid residues for substrate recognition.

(**A**) Superposition of substrate binding pocket amino acid residues in CsAlaDC and AtSerDC. The amino acid residues of CsAlaDC are shown in spring green, the amino acid residues of AtSerDC are shown in plum, with the substrate specificity-related amino acid residue highlighted in a red ellipse. (**B**) Active-site-lining amino acid residues of SDC homologs from Embryophyta were identified. The height of each amino acid is scaled proportionally to the amount of information content (measured in bits). The first line depicts the conserved motif in all SerDC homologs from Embryophyta, whereas lines 2-5 represent the conserved motifs based on the variable third amino acid residue. (**C**) Histogram showing the distribution of the number of key motifs. (**D**) Histogram showing the number of key motifs in different plant orders. (**E**) Relative enzyme activities of wild-type CsAlaDC and mutant protein CsAlaDC^{F106Y} against Ala substrate (columns 1 and 2), and enzyme activities of wild-type AtSerDC and various AtSerDC mutant proteins against Ala substrate (columns 3-9) are presented. The percentage graph shows the relative activity of each protein compared to wild-type CsAlaDC activity (taken as a 100% benchmark). (**F**) Relative enzyme activities of wild-type AtSerDC and AtSerDC mutant proteins (columns 1-7) against Ser substrates, and enzyme activities of wild-type CsAlaDC and mutant protein CsAlaDC^{F106Y} against Ser substrates(columns 8, 9) were measured. The percentage graph shows the relative activity of each protein compared to the wild-type AtSerDC (taken as a 100% benchmark). Three independent experiments were conducted. (**G**) The EA contents of AtSerDC and its mutant AtSerDC^{Y111F} in *N. benthamiana*. (**H**) The EA contents of CsAlaDC and its mutant CsAlaDC^{F106Y} in *N. benthamiana*. The significance of the difference (P<0.05) was labeled with different letters according to Duncan's multiple range test.

(Supplementary figure 4), indicating that they had been successfully overexpressed in tobacco. Here, we found that a high level of EA was produced in the CsAlaDC-expressing tobacco leaves; while no EA product was detected in tobacco leaves infiltrated with the mutant CsAlaDC^{F106Y}. In addition, we did not detect EA products in the AtSerDC-expressing tobacco leaves, while a high level of EA was detected in tobacco leaves infiltrated with mutant AtSerDC^{Y111F}. As anticipated, EA was not detected in tobacco leaves infiltrated with EV (**Figure 4G and H**). These results further verified the critical role of Phe^{106} in the selective binding of alanine for CsAlaDC.

Key amino acids for the evolution of CsAlaDC enzymatic activity

CsAlaDC and AtSerDC have a high sequence similarity of 74.5% and a nearly identical structure with an RMSD (root mean square deviation) of only 0.77Å for monomer structures. However, the two enzymes catalyze different amino acid decarboxylation reactions, and AtSerDC exhibits significantly higher *Vmax* than CsAlaDC (**Figure 2B and C**). In light of this observation, we postulated a hypothesis: EA, generated via Ala decarboxylation by CsAlaDC, can be toxic and harmful to plants if accumulated excessively. Thus, during the evolution of plant serine decarboxylase into alanine decarboxylase, the enzyme has evolved not just to alter the substrate preference but also to reduce catalytic activity to control EA production within a suitable range. Based on this hypothesis, we suggest that mutating specific amino acids in CsAlaDC to those corresponding amino acids in SerDCs could enhance its activity, and the results could provide insights into the evolution of CsAlaDC enzymatic activity.

Dimerization is essential to AlaDC/SerDC activities because the active site is composed of residues from two monomers. To identify the amino acids repressing the enzymatic activity of CsAlaDC during evolution from SerDC, we analyzed the crystal structures and the amino acids at the dimer interface between CsAlaDC and AtSerDC. This analysis revealed that the amino acids at the dimer interface at positions 66, 97, 110, 114, 116, 117, 122, 315 and 345 are different in CsAlaDC and SerDCs (**Figure 1B**). Therefore, we mutated these amino acids of CsAlaDC into those in the corresponding positions of AtSerDC and CsSerDC, and performed enzyme activity assays (**Figure 5A**).

The results demonstrated that CsAlaDC^{L110F} and CsAlaDC^{P114A} exhibited significantly enhanced enzyme activity compared to the wild-type CsAlaDC, with a 2.1-fold and 1.59-fold increase, respectively. Furthermore, the catalytic activity of the CsAlaDC$^{L110F/P114A}$ double mutant exhibits a remarkable 2.3-fold increase compared to that of the wild-type protein (**Figure 5B**). These findings suggested a critical role of Leu^{110} and Pro^{114} of CsAlaDC in the evolution of enzymatic activity and provided a basis to improve CsAlaDC activity. It is possible that these amino acid residues could potentially augment the hydrophobic nature of the protein dimer interface.

In Vitro synthesis of theanine

The biosynthetic pathway of theanine in tea plants comprises two consecutive enzymatic steps: alanine decarboxylase facilitates the decarboxylation of alanine to generate EA, while theanine synthetase catalyzes the condensation reaction between EA and Glu to synthesize theanine. By conducting mutation screens, we discovered a CsAlaDC mutant protein (L110F/P114A) that exhibited a 2.3-fold higher catalytic activity compared to the wild-type protein. Subsequently, we employed an *in vitro* theanine synthesis system utilizing CsAlaDC and either glutamine synthetase (PsGS [P*seudomonas syringae* pv. syringae]) or gamma-glutamate methylamine ligase (MmGMAS [*Methylovorus mays*]) which have the ability to synthesize theanine from EA and Glu (30, 31).

The results illustrated the successful synthesis of theanine using CsAlaDC in conjunction with two theanine synthetases, employing Ala and Glu as substrates (**Figure 5C**). The theanine content generated by the combination of CsAlaDC^{L110F} and MmGMAS, as well as CsAlaDC$^{L110F/P114A}$ and MmGMAS, was 4.57-fold and 6.72-fold higher than the content produced in the combination of wild-type CsAlaDC and MmGMAS, respectively. Similarly, when combined with the PsGS,

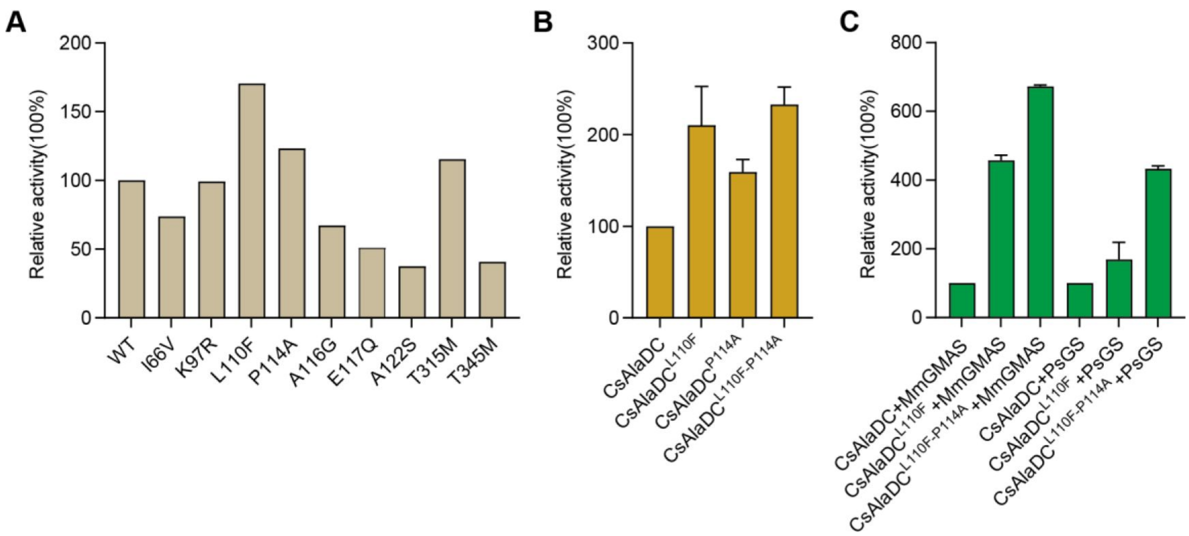

Figure 5.

Mutations enhance CsAlaDC enzyme activity and theanine synthesis *in vitro*.

(**A**) Relative enzyme activities of CsAlaDC mutant proteins against Ala substrate. (**B**) Relative enzyme activities of CsAlaDC^{L110F}, CsAlaDC^{P114A}, and CsAlaDC$^{L110F/P114A}$ against Ala substrate. (**C**) Histogram showing the relative content of theanine resulting from different combinations of alanine decarboxylase and theanine synthetase. Three independent experiments were conducted.

comparable outcomes are observed as well. The theanine content resulting from the combination of CsAlaDC^{L110F} and PsGS, as well as CsAlaDC$^{L110F/P114A}$ and PsGS, exhibit enhancements of 1.62-fold and 4.33-fold compared to the wild-type protein combination, respectively (**Figure 5C**). Hence, the utilization of CsAlaDC$^{L110F/P114A}$ could effectively enhance the theanine production yield and thus, holds potential for large-scale engineering production of theanine.

Discussion

The common and distinctive structural characteristics of CsAlaDC and AtSerDC compared with other amino acid decarboxylases

CsAlaDC and AtSerDC are Group II PLP-dependent amino acid decarboxylases, which exhibit numerous structural features common to other amino acid decarboxylases. The Dali search (32) revealed that AtSerDC and CsAlaDC share structural similarities with 7CIG (MetDC [*Streptomyces sp*. 590]), 7ERV (HisDC1 [*Photobacterium phosphoreum*]), 6KHO (TrpDC [*Oryza sativa Japonica* Group]), 4E1O (HisDC2 [*Homo sapiens*]), 6JY1 (AspDC [*Methanocaldococcus jannaschii*]), and 5GP4 (GluDC [*Levilactobacillus brevis*]). The monomers of these amino acid decarboxylases all consist of three characteristic domains: the C-terminal domain, the large domain and the N-terminal domain (Supplementary figure 5). The amino acid residues that bind to PLP cofactors are conserved across multiple enzymes (Supplementary figure 5 and 6). Out of the eight proteins analyzed, a total of thirteen amino acid residues were found to be conserved across all sequences (Glu^{143}, Glu^{171}, Lys^{201}, Gly^{245}, Thr^{247}, Asp^{253}, His^{275}, Asp^{277}, Ala^{279}, Pro^{286}, Ser^{302}, Lys^{309}, and Tyr^{336} in CsAlaDC). Some of them are situated within the active center and play a role in stabilizing PLP or facilitating catalytic reactions. Conversely, other residues reside outside the active center and their function remains unclear (Supplementary figure 6).

Group II PLP-dependent amino acid decarboxylases are characterized by the existence of a highly flexible loop, which is of great significance for the catalytic mechanism of decarboxylase (33). The loops are located at the dimer interface and extend to the active sites of other monomers in a closed conformation. Prior research has established that the conserved amino acid residue Tyr within the loop plays a crucial role in catalysis by donating protons to carbanions of quinone intermediates that arise following decarboxylation (34, 35). Our experiments demonstrate that the substitution of the corresponding Tyr with Phe in the loop renders the protein inactive. Through the application of circular dichroism, we have corroborated that the stability of the protein is not compromised by mutations (Supplementary figure 7A, B, D, E). Both the mutant and wild-type proteins manifest absorption bands at the 420 nm wavelength, signifying the formation of a Schiff base between PLP and the lysine residues found at the active site. Given that PLP has been incorporated during the protein purification stage, the absorbance observed for both the mutant and wild-type proteins at 420 nm is consistent when compared at equivalent concentrations (Supplementary figure 8). This corroboration implies that the protein mutation does not interfere with the binding to PLP. Consequently, the inactivation observed in the mutant proteins is not a consequence of instability incited by corresponding mutations, nor changes in the binding affinity between the protein and PLP induced by these mutations.

We observed that the substitution of Phe^{106} for Tyr in CsAlaDC rendered CsAlaDC inactive, while the substitution of Tyr^{111} for Phe in AtSerDC enabled alanine decarboxylase activity, and the alteration in the activity of the mutant protein bears no correlation to protein stability and affinity for PLP (Supplementary figure 7A, C, D, F and Supplementary figure 8), suggesting that the amino acid in the position is critical for determining substrate specificity. Integrative crystallographic structure of the two enzymes, we speculate the hydrophilic nature of the Lys residue in AtSerDC predisposes the active site to preferentially accommodate the hydrophilic amino acid Ser as its substrate. In contrast, the equivalent position in CsAlaDC is occupied by Phe, an amino acid lacking the hydroxyl group. This substitution enhances the hydrophobic nature of the substrate-binding pocket. Consequently, CsAlaDC demonstrates a unique predilection, selectively binding Ala (an amino acid with comparatively hydrophobic properties) as its preferred substrate.

The monomeric configuration of CsAlaDC and AtSerDC is akin to that of other Group II PLP-dependent amino acid decarboxylases, except for a conspicuous dissimilarity. Unlike other Group II PLP-dependent amino acid decarboxylases, the C-terminal of CsAlaDC and AtSerDC harbors an evident zinc finger structure. We propose that this structure could potentially influence enzyme stability. To test this hypothesis, we truncated the zinc finger structure from both proteins and expressed them. Our findings indicate that, following the excision of the zinc finger structure, CsAlaDC became insoluble, while AtSerDC displayed a similar trait to a certain extent (Supplementary figure 9). Hence, we can establish that the zinc finger structure indeed exerts an influence on the enzyme's stability. The potential for additional functions necessitates further investigation.

Evolution of AlaDC and SerDC

The structure of CsAlaDC seems like with AtSerDC, however, there are differences in substrate specificity. To clarify the relationship between CsAlaDC and other SerDCs, we analyzed serine decarboxylase-like proteins in Embryophyta that are highly homologous to CsAlaDC and constructed phylogenetic trees to gain insight into their evolutionary relationships (Supplementary figure 3). Based on our experimental findings and evolutionary evidence, we have identified a conserved YPX motif at the substrate binding pocket. For most plants, the residue X in this motif is predominantly Tyr, which corresponds to the serine decarboxylase protein. However, other residues such as Thr, Phe, Val, Ala, and Ile can also occupy this position (**Figure 4B and C**). Interestingly, serine decarboxylase-like proteins containing YPF motifs, which have the potential to function as alanine decarboxylases, were found to be distributed throughout the phylogenetic tree, including Asteroideae, Ericales, Chenopodiaceae, Poaceae, and Ranunculales (Supplementary figure 3B). However, these proteins were absent in some more recent species. This observation implies that the emergence of alanine decarboxylase in these particular species could be attributed to convergent evolution.

Moreover, we have observed the enrichment of serine decarboxylase-like proteins that possess a YPT motif within the Fabales (Supplementary figure 3B). Prior research has demonstrated that XP_004496485-1, which possesses a YPT motif, displays catalytic efficacy toward the processes of decarboxylation and oxidative deamination of Phe, Met, Leu, and Trp (36). Given that the YPT motif is highly conserved and widely distributed in Fabales, serine decarboxylase-like proteins bearing the YPT motif may have developed a unique substrate specificity in Fabales, beyond their conventional decarboxylation functions, as exemplified by XP_004496485-1 protein mentioned above. We speculate that these proteins may be capable of catalyzing other reactions, potentially involving non-protein amino acids or other substances as substrates.

Applied to improve the synthesis of theanine

Theanine is an important indicator of green tea quality. Therefore, improving the synthesis of theanine in tea plants is the focus of research. In this study, through crystal structure analysis and mutation verification, we found that the catalytic activity of CsAlaDC$^{L110F/P114A}$ is 2.3 times higher than that of the wild-type protein, resulting in a more abundant synthesis of theanine *in vitro*. This gives us great inspiration to improve the synthesis of EA in tea plants by gene editing, thus increasing the content of theanine in tea plants (**Figure 5**).

Theanine is highly demanded, by the market, due to its health effects and medicinal value, and as a constituent in food, cosmetics and other fields. To meet the market demand, a variety of methods have been used to acquire theanine, with the main methods including direct extraction, chemical synthesis, biotransformation (microbial fermentation) and plant cell culture. Based on the findings of this study, site-directed mutagenesis can be employed to modify enzymes involved in theanine synthesis. This modification enhances the capacity of bacteria, yeast, model plants, and other organisms to synthesize theanine, thereby facilitating its application in industrial theanine production.

Conclusions

In conclusion, our structural and functional analyses have significantly advanced understanding of the substrate-specific activities of alanine and serine decarboxylases, typified by CsAlaDC and AtSerDC. Critical amino acid residues responsible for substrate selection were identified-Tyr111 in AtSerDC and Phe106 in CsAlaDC-highlighting pivotal roles in enzyme specificity. The engineered CsAlaDC mutant (L110F/P114A) not only displayed enhanced catalytic efficiency but also substantially improved L-theanine yield in a synthetic biosynthesis setup with PsGS or MmGMAS. Our research expanded the repertoire of potential alanine decarboxylases through the discovery of 13 homologous enzyme candidates across embryophytic species and uncovered a special motif present in serine protease-like proteins within Fabale, suggesting a potential divergence in substrate specificity and catalytic functions. These insights lay the groundwork for the development of industrial biocatalytic processes, promising to elevate the production of L-theanine and supporting innovation within the tea industry.

Materials and Methods

Plants materials

Tobacco (*Nicotiana benthamiana*) plants were grown in a controlled chamber, under a 16-h light and 8-h dark photoperiod at 25 °C. Leaves of 5-week-old tobacco plants were used for transient transformation, mediated by *Agrobacterium tumefaciens* strain GV3101.

Gene cloning and protein expression

The coding sequences of AtSerDC and CsAlaDC, amino acid residues 66-482 and 61-478, respectively, were amplified using PCR and then ligated into the *Nde* I and *Xho* I restriction sites of the pET-28a and pET-22b vector, respectively, providing the recombinant vector pET-28a-*AtSerDC* and pET-22b-*CsAlaDC*. The cDNAs encoding CsSerDC were amplified using PCR and then ligated into the *Nde* I and *Xho* I restriction sites of the pET-28a vector. Gene-specific primers were listed in Supplementary table 2. The recombinant plasmid was transformed into *E. coli* BL21 (DE3) competent cells. Positive transformants were grown in a 5 mL LB medium containing 30 μg/mL kanamycin or 50 μg/mL ampicillin at 37 °C overnight and then subcultured into an 800 mL LB medium containing the corresponding antibiotic. Protein expression was induced by the addition of 0.2 mM isopropyl-β-d-thiogalactoside (IPTG) for 20 h at 16 °C when the optical density (OD) at 600 nm reached 0.6-0.8, harvested by centrifugation at 4 °C and 4,000 rpm for 30 min.

Protein production and crystallization

The cell pellet was suspended in 30 mL of lysis buffer (20 mM HEPES, pH 7.5, 200 mM NaCl, 0.1 mM PLP), disrupted by High-Pressure Homogenizer, and then centrifuged at 16,000 rpm for 30 min at 4 °C to remove the cell debris. The supernatants were purified with a Ni-Agarose resin column followed by size-exclusion chromatography. Before crystallization, purified proteins were concentrated at 10 mg/mL. Crystallization conditions were screened by the sitting-drop vapor diffusion method using the reservoir solutions supplied in commercially available screening kits (Crystal Screen, Crystal Screen 2, PEGRx 1, 2, and SaltRx 1, 2). A droplet made by mixing 1.0 μL of purified AtSerDC or CsAlaDC (10 mg/mL) with an equal volume of a reservoir solution was equilibrated against 100 µL of the reservoir solution at 16℃. The crystal of AtSerDC was obtained using buffer pH 8.0 containing 20% (w/v) PEG400 as a precipitate and 0.2 M $CaCl_2$. The crystal of the CsAlaDC-EA complex was obtained at pH 7.5 containing 2.6 M sodium acetate. The crystal of CsAlaDC was obtained at pH 6.0 containing 3.5 M sodium formate.

Data collection and processing

Crystals were grown by sitting-drop vapor diffusion method at 16 °C. The volume of the reservoir solution was 100 µL and the drop volume was 2 µL, containing 1 µL of protein sample and 1 µL of reservoir solution. The reservoir solution of AtSerDC contained 0.1 M Tris-HCl (pH 8.0), 0.2 M $CaCl_2$, and 20% PEG 400. The reservoir solution of CsAlaDC contained 0.1 M HEPES (pH 7.5), 2.6 M sodium acetate or 0.1 M Bis-Tris (pH 6.0), and 3.5 M sodium formate. Crystals grew in 3-5 days using a protein concentration of 10 mg/mL. Diffraction data were collected at Shanghai Synchrotron Radiation Facility (China). The collected data sets were indexed, integrated, and scaled using the HKL3000 software package. The structure of AtSerDC was solved by molecular replacement using the structure of HisDC (PDB code: 7ERV [*Photobacterium phosphoreum*]) as the model, and utilizing AtSerDC as a molecular displacement model for both CsAlaDC and the CsAlaDC-EA complex. The AtSerDC, CsAlaDC, and CsAlaDC-EA complex exhibit resolutions of 2.85 Å, 2.50 Å, and 2.60 Å, respectively. The statistics for data collection and processing are summarized in Supplementary table 1.

Enzyme activity assays

Decarboxylase activity was measured by detecting products (EA or ethanolamine) in Waters Acquit ultraperformance liquid chromatography (UPLC) system (16, 37). The 100 µL reaction mixture, containing 20 mM substrate (Ala or Ser), 100 mM potassium phosphate, 0.1 mM PLP, and 0.025 mM purified enzyme, was prepared and incubated at standard conditions (45 °C and pH 8.0 for CsAlaDC, 40 °C and pH 8.0 for AtSerDC for 10 min). Then, the reaction was stopped with 20 µL of 10% trichloroacetic acid. The product was derivatized with 6-aminoquinolyl-N-hydroxy-succinimidyl carbamate (AQC) and subjected to analysis by UPLC. All enzymatic assays were performed in triplicate.

The detection methodology for theanine production remains consistent with the aforementioned approach. The 100 µL reaction mixture, containing 20 mM Ala, 45 mM Glu, 50 mM HEPES (pH 7.5), 0.1 mM PLP, 30 mM $MgCl_2$, 10 mM ATP, 0.03 mM PsGS/MmGMAS, and 0.025 mM CsAlaDC/CsAlaDC ^{L110F}/CsAlaDC $^{L110F/P114A}$, was prepared and incubated at standard conditions for 1 h. Subsequently, the reaction was terminated via immersion of the reaction vessel in a metal bath at 96 °C for 3 min (31). Theanine was derivatized with AQC and subjected to analysis by UPLC. All enzymatic assays were performed in triplicate.

Site-directed mutagenesis

Site-directed mutagenesis experiment was conducted using a PCR method from the wild-type construct pET-28a-*AtSerDC* and pET-22b-*CsAlaDC*, respectively. *Dpn* I endonucleases were used to digest the parental DNA template. The reaction mixture was used to transform *E. coli* DH5α competent cells and the plasmids from positive strains were extracted to *E. coli* BL21 (DE3) for protein expression, purification, and analysis of the enzymatic activity.

In vivo enzyme activity assay in *N. benthamiana*

The amplified PCR products were fused to the plant expression vector, pCAMBIA1305. Linearization was conducted by restriction digest with *Spe* I and *Bam*H I. The recombinant colonies were selected for PCR validation on the appropriate antibiotics plate. After validation, the plasmids were electroporated into *Agrobacterium tumefaciens* strain GV3101(pSoup-p19). Empty pCAMBIA1305 vector, with an intron containing the GFP gene, was treated as the control.

Agrobacterium transient expression assays were performed on 5-week-old *N. benthamiana* plants. *Agrobacterium tumefaciens* strain GV3101 (pSoup-p19), carrying the above-described vectors, were cultured in Luria Bertani (LB) medium, containing appropriate antibiotics, at 28 °C. When the absorbance of bacteria colonies reached OD_{600} = 0.6-0.8, bacterial cells were collected and resuspended in MMA solution (10 mM $MgCl_2$, 10 mM 2-(N-morpholino) ethane sulfonic acid (MES), pH 5.6). After the OD_{600} of the resuspended bacterial solution was adjusted to approx. 1.0, acetosyringone (AS) was added with a final concentration of 200 μM, and this solution was then incubated, at room temperature, for at least 3 h in darkness. Next, cell suspensions were infiltrated into *N. benthamiana* leaves with a needle-free syringe. These *N. benthamiana* leaves were then collected 3 days post infiltration, frozen in liquid nitrogen and stored at –80 °C. Internal EA in *N. benthamiana* leaves was extracted as previously described (18), and then the solvent was subjected to gas chromatography-mass spectrometry GS-MS system.

Transcript level analysis in *N. benthamiana*

Total RNA was isolated from samples using the RNAprep Pure Plant Kit (Tiangen, Beijing, China), according to the manufacturer's protocol. The cDNAs were synthesized using TransScript One-Step gDNA Removal and cDNA Synthesis SuperMix Kit (TransGen Biotech, Beijing, China). The qRT-PCR assays were performed, as previously described (38). Primers used for qRT-PCR assays were listed in Supplementary table 3 and the qRT-PCR was run on a Bio-Rad CFX96TM RT PCR detection system and CFX Manager Software. Each reaction reagent (20 μL) contained 0.4 μL forward and reverse primers (10 μM), 2 μL cDNA (200±5 ng/μL), 10 μL SYBR Green Supermix (Vazyme, Nanjing, China) and 6.2 μL double-distilled water. Reaction was performed by a two-step method: 95 ℃ for 5 min; 40 cycles of 95 ℃ for 10 s; and 60 ℃ for 30 s. The glyceraldehyde-3-phosphate dehydrogenase (GAPDH) gene was used for internal normalization in each RT-PCR, and the $2^{-\Delta\Delta CT}$ method (39) was used to calculate the relative gene expression. All samples were performed with three replicates.

Multiple sequence alignment

MUSCLE was used to generate the protein multiple sequence alignment of amino acid decarboxylases with default settings (40). ESPript 3.x was used to display the multiple sequence alignment (41).

Phylogenetic analysis

The AtSerDC was used as a query for searching its homologs in Embryophyta using BLASTP (42) (version 2.13.0). To fetch homologs of AtSerDC in distant species, we performed a two-round blast. We first search for the best hit in the given order under Embryophyta. Then we blast each species under a provided order using the order best hit as a query and as many as 10 hits were kept. The sequences with e values <= 0.001 were removed. We further dropped sequences with lengths less than 200 or larger than 800 amino acids. Then multiple sequence alignment was performed via clustalo (version 1.2.4) with default settings on these filtered sequences (43). The NJ algorithm implemented in clustalw (version 2.1) was adopted to build the phylogenetic tree (44). The R package ggtree (version 3.2.1) and universalmotif (version 1.12.4) were used for tree and motif visualization (45).

Author contributions

W.G, Z.Z and X.W designed research; H.W, B.Z, S.Q and C.D performed research; H.W and B.Z analyzed data; H.W, B.Z, W.G, and Z.Z wrote the paper.

Competing interest statement

The authors declare no competing financial interests.

Classification

Biochemistry (major), Plant Sciences (minor)

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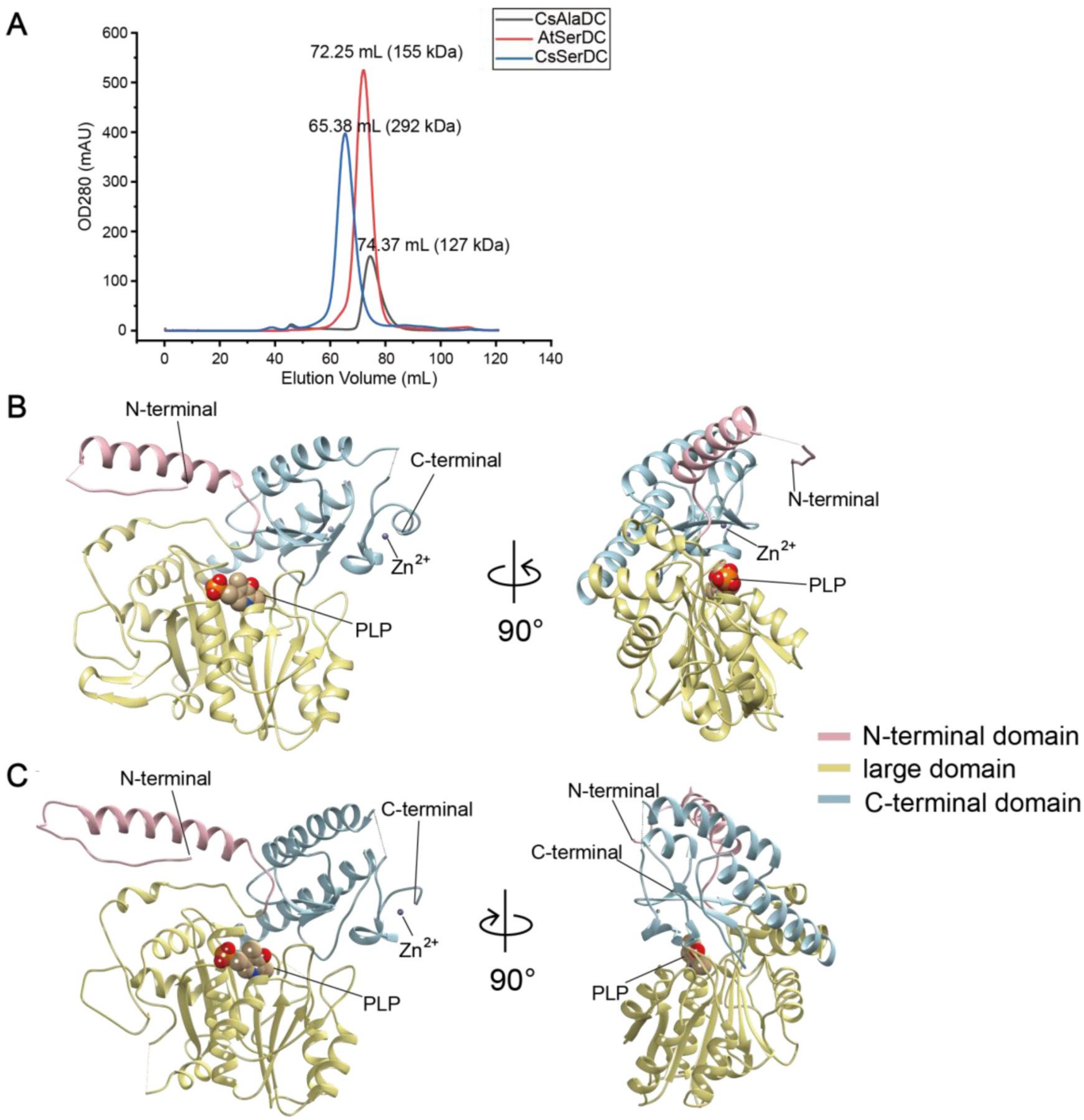
Data availability

The structures of AtSerDC, CsAlaDC-EA complex and CsAlaDC have been deposited in the Protein Data Bank (PDB) under accession numbers 8JG7, 8JIK and 8JIJ respectively.

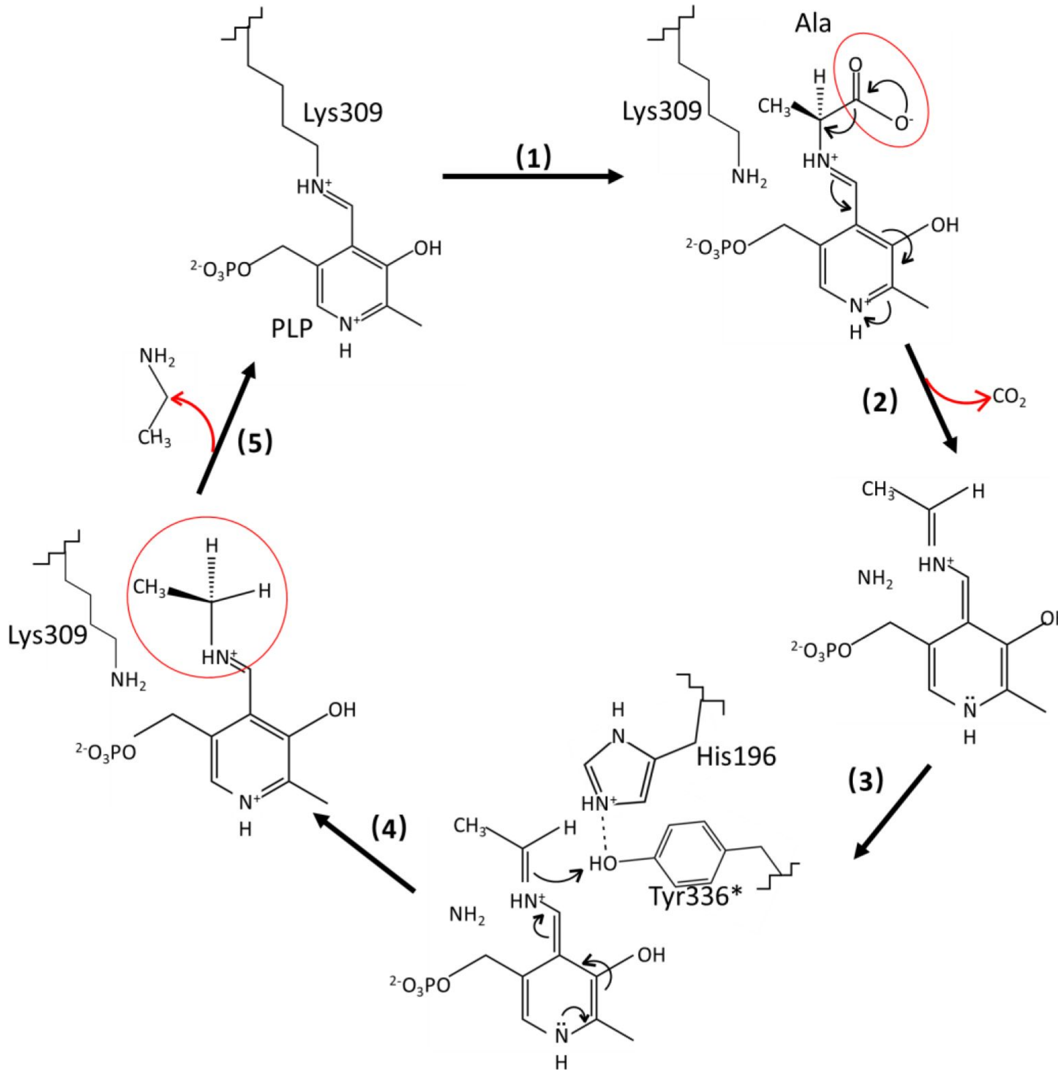
Supplementary figures



Supplementary figure 1.

Purification of CsAlaDC, AtSerDC, and CsSerDC and Crystal structures of CsAlaDC and AtSerDC.

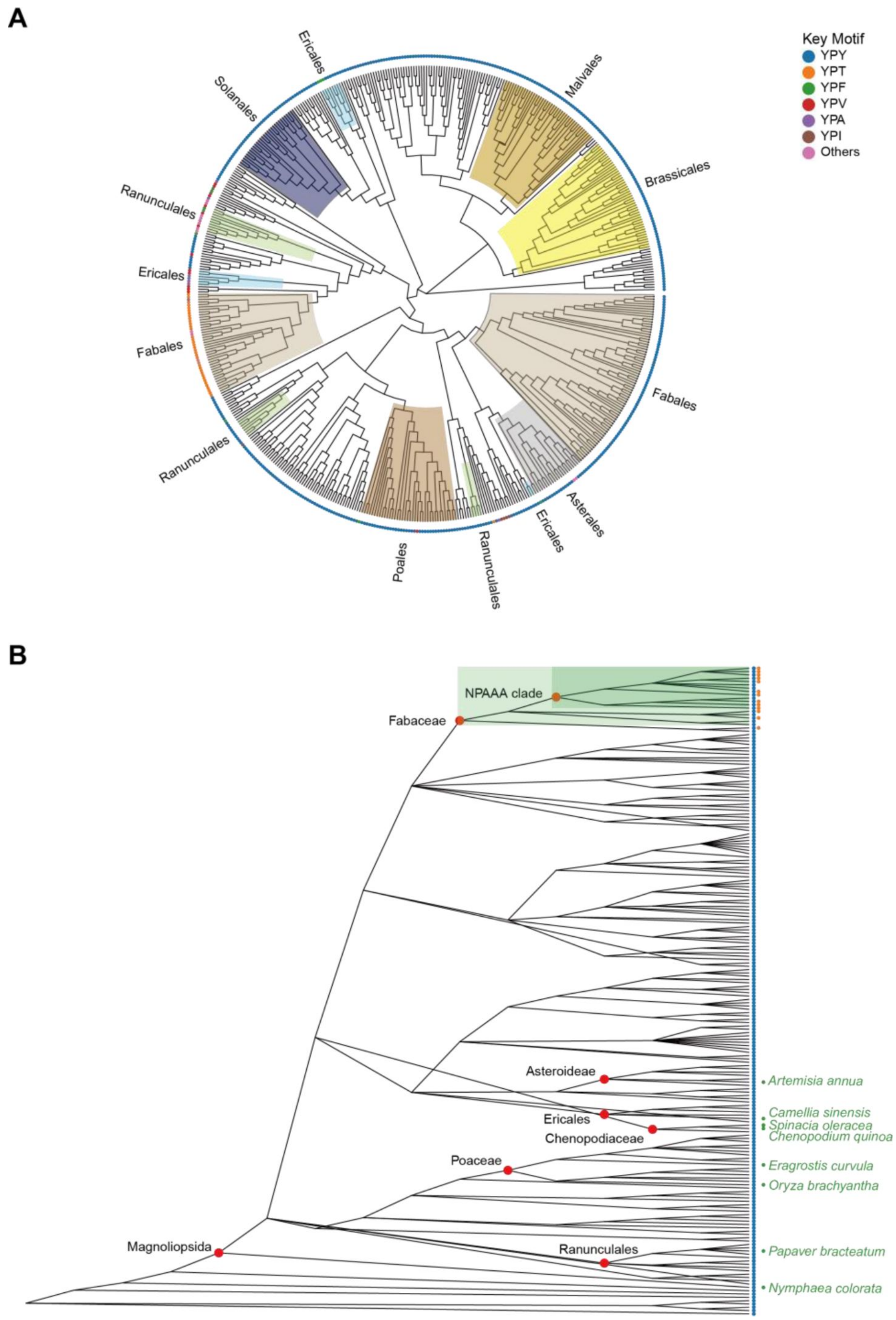
(**A**) Comparison of elution profiles of CsAlaDC, AtSerDC and CsSerDC. (**B**) Monomer structure of CsAlaDC. (**C**) Monomer structure of AtSerDC. The color display of the N-terminal domain, large domain, and C-terminal domains is shown in light pink, khaki and sky blue, respectively. The PLP molecule is shown as a sphere model. The gray spheres represent zinc ions.



Supplementary figure 2.

Catalytic mechanisms and conformational changes of CsAlaDC.

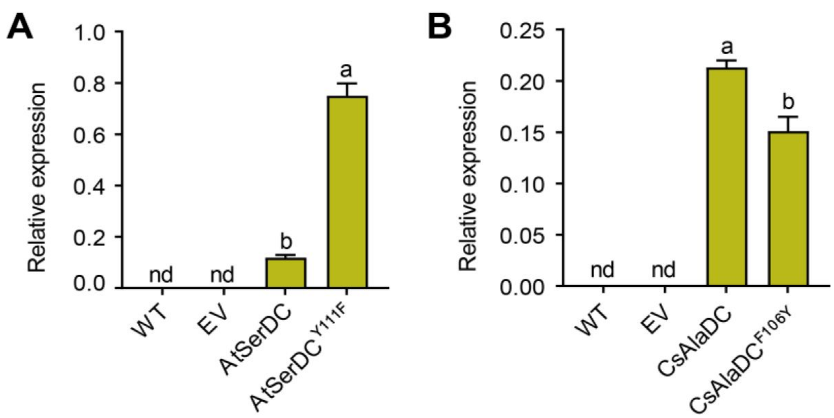
After the transaldimation of the internal aldimine within CsAlaDC, resulting in the release of the active-site residue Lys309, the PLP amino acid external aldimine undergoes decarboxylation, leading to the removal of the α-carboxyl group as CO2. This process generates a quinonoid intermediate that is stabilized by the delocalization of paired electrons (1,2). The carbanion at Cα is subsequently protonated by the acidic p-hydroxyl group of Tyr336* located on the large loop, facilitated by its neighboring residue His196, which is situated on the small loop (3,4). Simultaneously, the internal aldimine LLP309 in CsAlaDC is restored, resulting in the release of the product (5).



Supplementary figure 3.

Evolutionary analysis of CsAlaDC in Embryophyta.

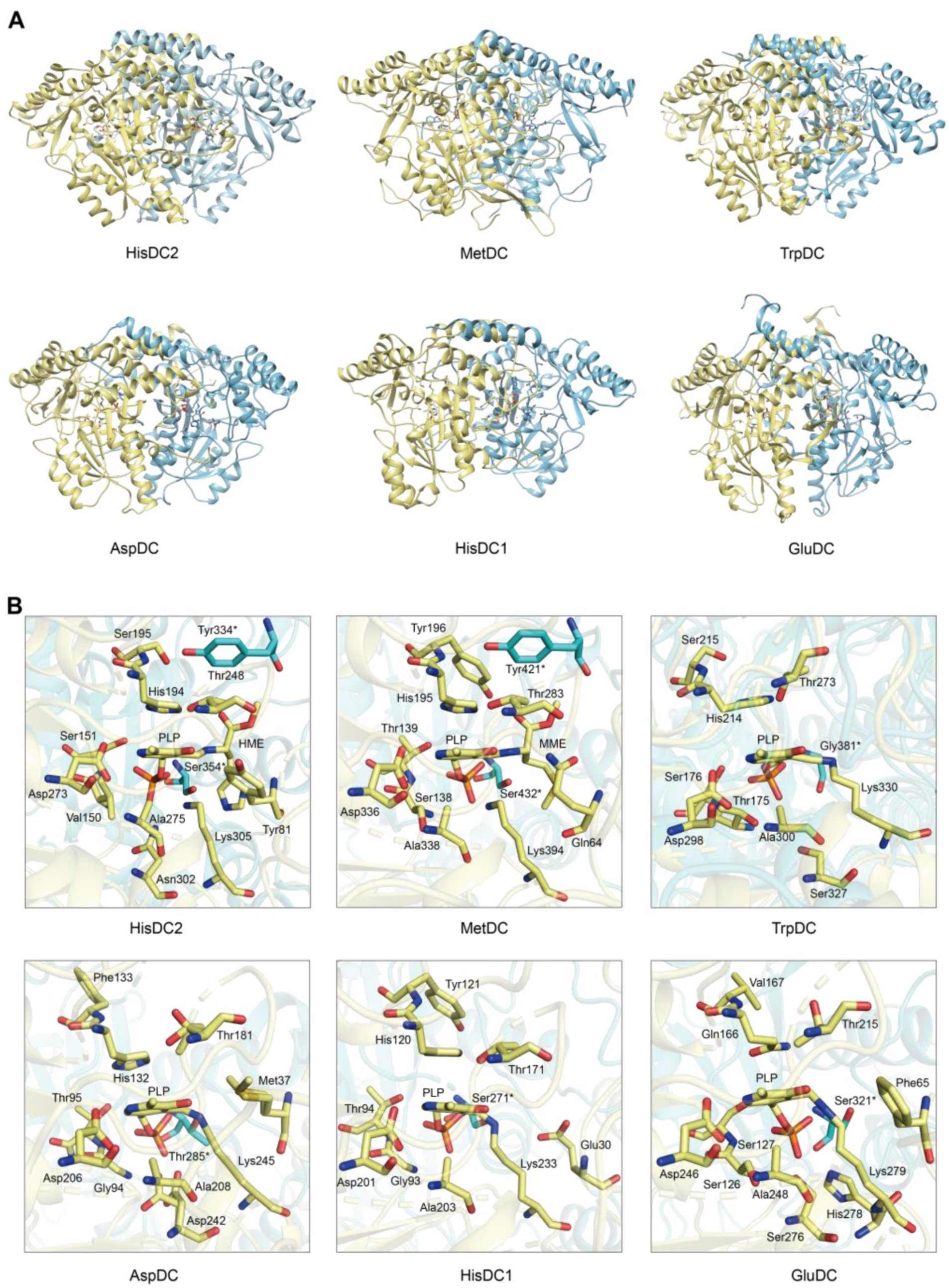
(**A**) The presented diagram depicts an evolutionary tree of CsAlaDC, which is devoid of a root and solely portrays the topological structure of the tree without including distance information. The color of the inner ring corresponds to various orders, while the outer ring's leaf nodes are colored based on the motif types that the sequence exhibits. (**B**) Diversity of serine decarboxylase-like proteins in Embryophyta (196 species). Colored scatter spots on the right side of leaf nodes correspond to the respective motifs shown in Figure A.



Supplementary figure 4.

The relative mRNA levels of AtSerDC and its mutant Y111F

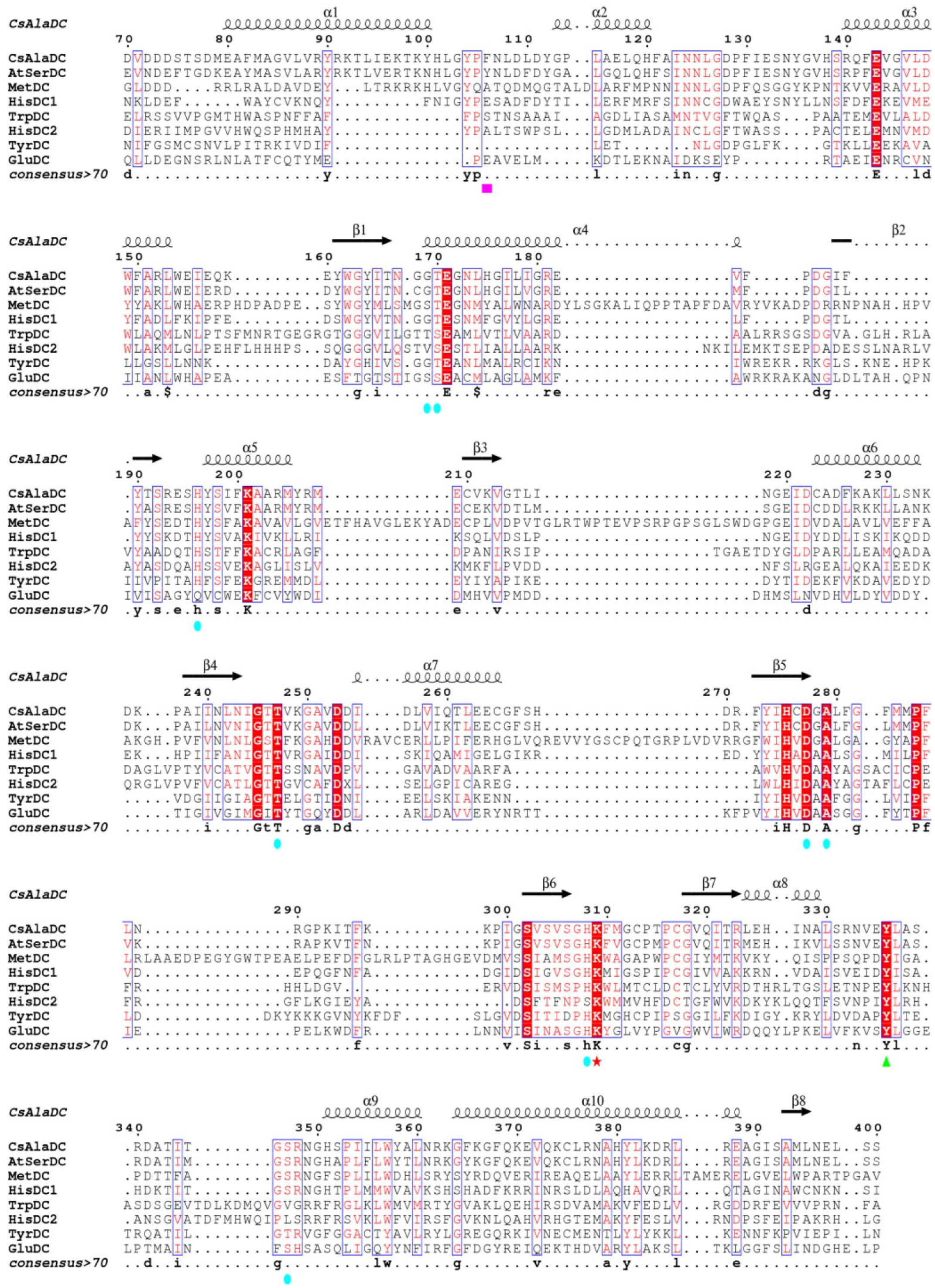
(*A*), CsAlaDC and its mutant F106Y (*B*) in *N. benthamiana* leaves were measured by two primers. WT, wild type of *N. benthamiana*; EV, empty vector control; *NbGAPDH* was used as an internal control. Data represent mean ± SD (n=3). The significance of the difference (P<0.05) was labeled with different letters according to Duncan's multiple range test.



Supplementary figure 5.

Structures of HisDC2, MetDC, TrpDC, AspDC, HisDC1 and GluDC.

(**A**) The Overall Structures of HisDC2, MetDC, TryDC, AspDC, HisDC1 and GluDC. Chain A is shown in khaki, chain B is shown in cyan. (**B**) Amino acid residues in the substrate binding pocket of HisDC2, MetDC, TryDC, AspDC, HisDC1 and GluDC. The amino acid residues in chain A are shown in khaki, and the amino acid residues in chain B are shown in cyan.



Supplementary figure 6.

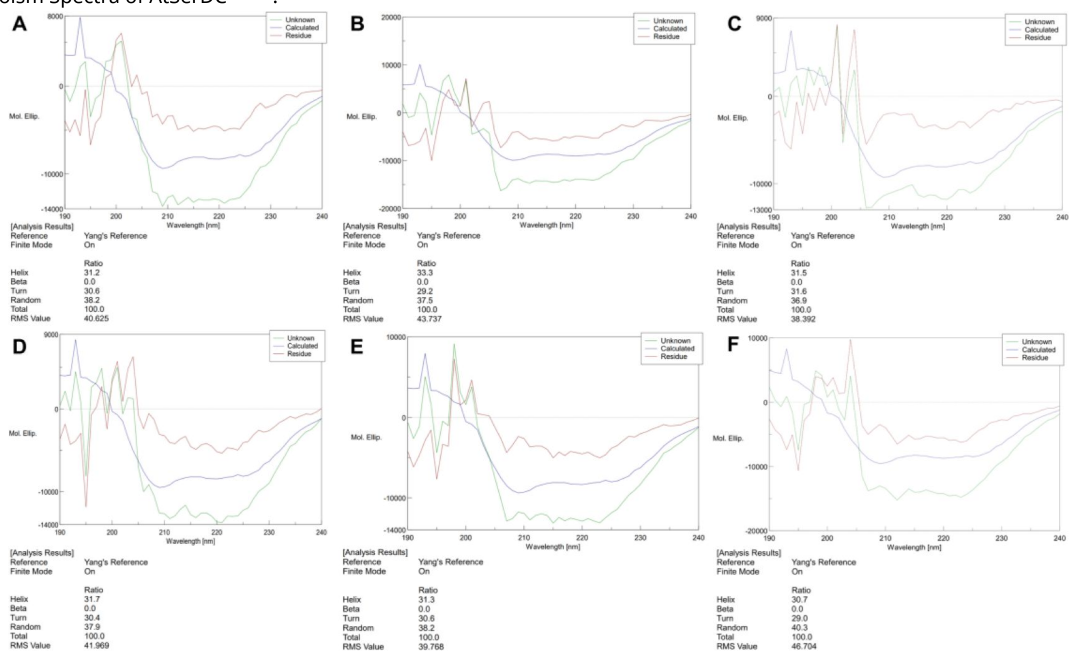
Multiple sequence alignment of CsAlaDC, AtSerDC, MetDC, HisDC1, TrpDC, HisDC2, TyrDC, and GluDC were generated using MUSCLE and visualized with ESPript 3.x. Conserved amino acid residues in all eight proteins are highlighted with red backgrounds.

The magenta box marks key amino acid residues involved in substrate recognition for CsAlaDC and AtSerDC. The Lys residue covalently bound to the PLP cofactor is denoted by a red star, while the green triangle indicates the Tyr residue associated with enzymatic activity. Amino acid residues involved in the CsAlaDC substrate binding pocket are marked with blue circles.

Supplementary figure 7.

Circular Dichroism Spectra of proteins.

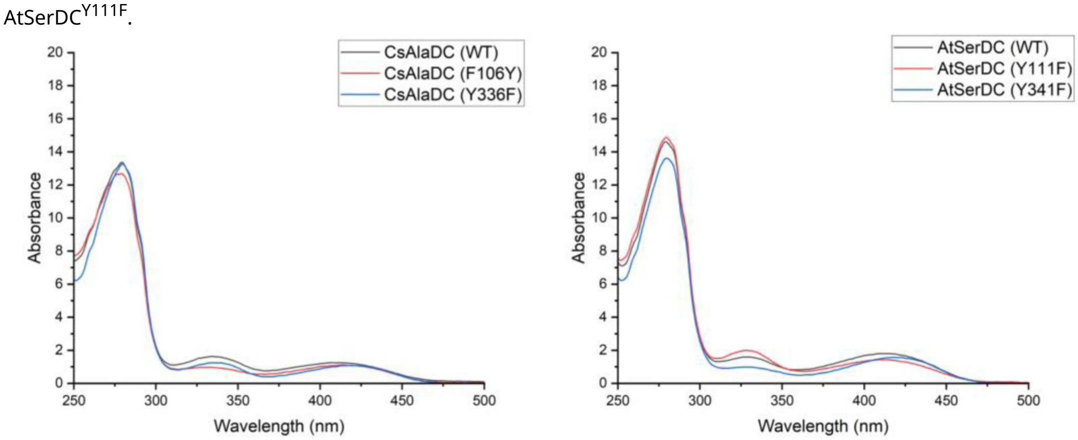
(**A**) Circular Dichroism Spectra of CsAlaDC (WT). (**B**) Circular Dichroism Spectra of CsAlaDC^{Y336F}. (**C**) Circular Dichroism Spectra of CsAlaDC^{F106Y}. (**D**) Circular Dichroism Spectra of AtSerDC (WT). (**E**) Circular Dichroism Spectra of AtSerDC^{Y341F}. (**F**) Circular Dichroism Spectra of AtSerDC^{Y111F}.

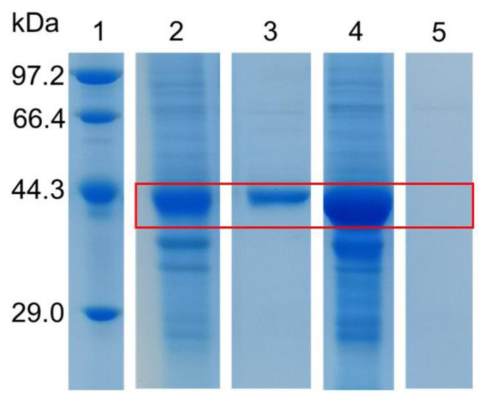


Supplementary figure 8.

Absorption Spectra of different proteins.

(**A**) Absorption Spectra of CsAlaDC (WT), CsAlaDC^{Y336F} and CsAlaD^{F106Y}. (**B**) Absorption Spectra of AtSerDC (WT), AtSerDC^{Y341F} and AtSerDC^{Y111F}.





Supplementary figure 9.

Purification of truncated CsAlaDC and AtSerDC.

Lane 1 represents marker. The precipitated and eluted samples of AtSerDC with a truncated zinc finger structure are shown in lanes 2 and 3 respectively, the precipitated and eluted samples of CsAlaDC with a truncated zinc finger structure are illustrated in lanes 4 and 5 respectively.

Supplementary tables

	CsAlaDC	CsAlaDC-EA complex	AtSerDC
Data Collection			
Space Group	*P3121*	*P3121*	*C2*
Cell Dimensions			
a, b, c (Å)	163.5,163.5, 50.2	164.0,164.0,50.2	115.4,149.5,139.1
α, β, γ (°)	α=β=90, γ=120	α=β=90, γ=120	α=γ=90, β=109.2
Wavelength (Å)	0.979	0.979	0.979
Resolution (Å)	50-2.50	50-2.60	50-2.85
Completeness (%)	99.3 (99.9)	99.6 (100.0)	99.6 (98.1)
Redundancy	5.4 (5.1)	13.0 (12.5)	5.9 (5.0)
$\langle I\rangle/\langle\sigma I\rangle$	9.3 (1.9)	18.9 (4.6)	10.6 (2)
R_{merge}	0.128 (0.603)	0.156 (0.562)	0.173 (0.551)
Refinement			
R_{work}/ R_{free}	0.173/0.222	0.172/0.223	0.214/0.262
No. reflections	25175	22813	49543
B-facator (Å^2)			
Protein	43.3	48.3	57.6
PLP	29.5	31.5	52.8
Zn^{2+}	58.7	43.8	71.6
Ca^{2+}	-	55.5	-
Ethylamine	-	27.5	-
Glycerol	-	-	51.7
Water	38.5	40.9	-
R.m.s. deviations			
Bond lengths (Å)	0.008	0.008	0.007
Bond angles (°)	1.531	1.601	1.410
No. of atoms			
Protein	3292	3265	3250
PLP	15	15	60
Zn^{2+}	2	1	4
Ca^{2+}	-	1	-
Ethylamine	-	5	-
Glycerol	-	-	24
Water	115	60	-
Ramachandran statistics (%)			
Favoured	94.01	93.58	91.52
Allowed	5.74	5.68	7.61
Outliers	0.25	0.74	0.87

R_{merge}=∑hkl∑i|Ii(hkl)-<I(hkl)>|/∑hkl∑i Ii(hkl), where<I(hkl)>is the main value of I(hkl).

R_{work}=∑||Fobs|-|Fcalc||/∑|Fobs|, where Fobs and Fcalc are observed and calculated structure factors.

The free R factor was calculated using 5% of reflections ommitted from the refinement.
Highest resolution shell is shown in parenthesis

Supplementary table 1.

Data collection and refinement statistics

Supplementary table 2.

Primers used for gene cloning.

Name	Forward primers (5'-3')	Reverse primers (5'-3')
CsAlaDC	GGAATTCCATATGACAACAAGCCTCACCATCACGG	GATCTCGAGTCATTTATGAAGATCACAATCACAATTCTCACTTC
AtSerDC	GGAATTCCATATGACCACCAGCCTGGCCG	GATCTCGAGTTATTTATGGGCCGGACAAATGCAATTATTG
CsSerDC	ATGGTGGGAAGTGTTGGGGT	TCACTTGTGTAGTGCACAAAGACAGT

Supplementary table 3.

Primers used for real-time PCR.

Name	Forward primers (5'-3')	Reverse primers (5'-3')
CsAlaDC	CACTGTGATGGGGCTCTGTT	TGTTATCTGGACACCGCACG
AtSerDC	TCACTCGCTGTAACGGAACC	AACTGACCAAGCGCACCATA
NbGAPDH	AGCTCAAGGGAATTCTCGATG	AACCTTAACCATGTCATCTCCC

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Editors

Reviewing Editor
Amie Boal
Pennsylvania State University, University Park, United States of America

Senior Editor
Amy Andreotti
Iowa State University, Ames, United States of America

Reviewer #3 (Public Review):

In the manuscript titled "Structure and Evolution of Alanine/Serine Decarboxylases and the Engineering of Theanine Production," Wang et al. solved and compared the crystal structures of Alanine Decarboxylase (AlaDC) from Camellia sinensis and Serine Decarboxylase (SerDC) from Arabidopsis thaliana. Based on this structural information, the authors conducted both in vitro and in vivo functional studies to compare enzyme activities using site-directed mutagenesis and subsequent evolutionary analyses. This research has the potential to enhance our understanding of amino acid decarboxylase evolution and the biosynthetic pathway of the plant specialized metabolite theanine, as well as to further its potential applications in the tea industry.

https://doi.org/10.7554/eLife.91046.3.sa2

Reviewer #2 (Public Review):

Summary:

The manuscript focuses on comparison of two PLP-dependent enzyme classes that perform amino acyl decarboxylations. The goal of the work is to understand the substrate specificity and factors that influence catalytic rate in an enzyme linked to theanine production in tea plants.

Strengths:

The work includes x-ray crystal structures of modest resolution of the enzymes of interest. These structures provide the basis for design of mutagenesis experiments to test hypotheses about substrate specificity and the factors that control catalytic rate. These ideas are tested via mutagenesis and activity assays, in some cases both in vitro and in plants.

Weaknesses:

Although improved in a revision, the manuscript could be more clear in explaining the contents of the x-ray structures and how the complexes studied relate to the reactant and product complexes. Some of the figures lack sufficient clarity and description. Some of the claims about the health benefits of tea are not well supported by literature citations.

https://doi.org/10.7554/eLife.91046.3.sa1

Author response:

The following is the authors’ response to the previous reviews.

Response to reviewer’s comments

> ***Reviewer #2 (Public Review):***
>
> *Summary:*
>
> *The manuscript focuses on comparison of two PLP-dependent enzyme classes that perform amino acyl decarboxylations. The goal of the work is to understand the substrate specificity and factors that influence catalytic rate in an enzyme linked to theanine production in tea plants.*
>
> *Strengths:*
>
> *The work includes x-ray crystal structures of modest resolution of the enzymes of interest. These structures provide the basis for design of mutagenesis experiments to test hypotheses about substrate specificity and the factors that control catalytic rate. These ideas are tested via mutagenesis and activity assays, in some cases both in vitro and in plants.*
>
> *Weaknesses:*
>
> *Although improved in a revision, the manuscript could be more clear in explaining the contents of the x-ray structures and how the complexes studied relate to the reactant and product complexes. The manuscript could also be more concise, with a discussion section that is largely redundant with the results and lacking in providing scholarly context from the literature to help the reader understand how the current findings fit in with work to characterize other PLP-dependent enzymes or protein engineering efforts. Some of the figures lack sufficient clarity and description. Some of the claims about the health benefits of tea are not well supported by literature citations.*

Thank you for your insightful comments on our manuscript and your recognition of the strengths of our study. We understand your concerns about the weaknesses mentioned, and we have addressed them appropriately in the revised manuscript. We acknowledge that the discussion section needs to be improved for conciseness and context. We have revised this part by removing the redundant content. We also acknowledge your comments concerning the clarity and description of some figures. We have revisited these figures and revised them, ensuring they are clear and adequately described. Lastly, concerning the claims about the health benefits of tea, we understand your concern about the lack of supporting citations. We ensure to back such claims with valid literature or, if necessary, omit these statements.

> ***Reviewer #2 (Recommendations For The Authors):***
>
> *(1) Line 21: Alanine Decarboxylase should not be capitalized.*

Thank you very much for your careful reading of the manuscript. We have corrected it in the revised manuscript.

> *(2) Line 31: Grammatical error. Also not clear what "evolution analysis" means here. Revise to "Structural comparisons led us to..."*

Thank you very much for your careful reading of the manuscript. We have corrected it in the revised manuscript.

> *(3) Line 34: Revise to "Combining a double mutant of CsAlaDC"*

Thank you very much for your careful reading of the manuscript. We have corrected it in the revised manuscript.

> *(4) Line 35: Change word order to "increased theanine production 672%"*

Thank you very much for your careful reading of the manuscript. We have corrected it in the revised manuscript.

> *(5) Line 37: meaning unclear. Revise to "provides a route to more efficient biosynthesis of theanine."*

Thank you very much for your careful reading of the manuscript. We have corrected it in the revised manuscript.

> *(6) Line 44: I'm not sure that the "health effects" of tea have been proven in placebo controlled studies. And the references provided (2-4 and 5) do not describe original research articles supporting these claims. I would suggest removing these statements from the introduction and at later points in the manuscript.*

Thank you for your thoughtful feedback and suggestions. Based on your suggestion, we have removed these statements: "The popularity of tea is determined by its favorable flavor and numerous health benefits (2-4). The flavor and health-beneficial effects of tea are conferred by the abundant secondary metabolites, including catechins, caffeine, theanine, volatiles, etc (5). " As for the subsequent statement: " It has also many health-promoting functions, including neuroprotective effects, enhancement of immune functions, and potential anti-obesity capabilities, among others. " the referenced literature cited can substantiate this conclusion.

> *(7) Line 58: insert "the" between provided and basis*

Thank you very much for your careful reading of the manuscript. We have corrected it in the revised manuscript.

> *(8) Line 100: Not clear what this phrase means, "As expected, CsSerDC was closer to AtSerDC" Please clarify - closer to what?*

We apologize for any confusion caused by the unclear phrasing. When referring to "CsSerDC was closer to AtSerDC," we intended to convey that CsSerDC exhibits a higher degree of sequence homology with AtSerDC than it does with the other enzymes evaluated in our investigation. However, a 1.29% difference between 86.21% and 84.92% in amino acid similarity is not statistically significant (Figure 1B and Supplementary table 1 in the original manuscript), we have deleted the relevant descriptions in the revised manuscript.

> *(9) Line 112: "were constructed into" makes no sense. It would be better to say the genes for the proteins of interest were inserted into the overexpression plasmid.*

Thank you very much for your careful reading of the manuscript. We have corrected it in the revised manuscript.

> *(10) Line 115: missing the word "the" between generated and recombinant*

Thank you very much for your careful reading of the manuscript. We have corrected it in the revised manuscript.

> *(11) Line 121: catalyze not catalyzed*

Thank you very much for your careful reading of the manuscript. We have corrected it in the revised manuscript.

> *(12) Lines 129 and 130: The reported Km values are really large - in the mM range. Do these values make sense in terms of the available concentrations of the substrates inside the cell?*

The content of alanine in tea plant roots ranges from 0.28 to 4.18 mg/g DW (Yu et al., 2021; Cheng et al., 2017). Correspondingly, the physiological concentration of alanine is 3.14 mM to 46.92 mM, in tea plant roots. The content of serine in plants ranges from 0.014 to 17.6 mg/g DW (Kumar et al., 2017). Correspondingly, the physiological concentration of serine is 0.13 mM to 167.48 mM in plants. Therefore, in this study, the Km values are within the range of available substrate concentrations inside the cell.

Yu, Y. *et al.* (2021) Glutamine synthetases play a vital role in high accumulation of theanine in tender shoots of albino tea germplasm "Huabai 1". *J. Agric. Food Chem.* 69 (46),13904-13915.

Cheng, S. *et al*. (2017) Studies on the biochemical formation pathway of the amino acid L-theanine in tea (Camellia sinensis) and other plants. *J. Agric. Food Chem.* 65 (33), 7210-7216.

Kumar, V. *et al*. (2017) Differential distribution of amino acids in plants. Amino Acids. 49(5), 821-869.

> *(13) Line 211: it is unclear what the phrase "as opposed to wild-type" means. Please clarify.*

Thank you very much for your careful reading of the manuscript and valuable suggestions. We intend to communicate that the wild-type CsAlaDC and AtSerDC demonstrate decarboxylase activity, while the mutated proteins have experienced a loss of decarboxylation activity. We have already modified this concern in the revised version of the manuscript.

> *(14) Line 222: residues not residue*

Thank you very much for your careful reading of the manuscript. We have corrected it in the revised manuscript.

> *(15) Line 227 and Figure 4B: It is not clear what the different sequence logos mean in this part of the figure. The caption is too brief and not helpful. And the sentences describing this figure panel are also not sufficiently clear.*

Thank you very much for your careful reading of the manuscript and valuable suggestions. We have provided a more detailed explanation of this section in the revised manuscript and added additional annotations in the figure caption to provide further clarity.

> *(16) Lines 233 and 234: "in the substrate specificity" is awkwardly worded. I would revise to "in selective binding of the appropriate substrate."*

Thank you very much for your careful reading of the manuscript and valuable suggestions. We have meticulously revised the description of this section.

> *(17) Line 243: a word is missing in this sentence - but I can't figure out the intended meaning or what the missing word is. Rephrase to improve clarity.*

Thank you very much for your careful reading of the manuscript and valuable suggestions. We have revised this sentence to: " These findings indicate the essential role of Phe106 in the selective binding of alanine for CsAlaDC. "

> *(18) Line 255: The "expression system...was carried out" is not correct. I would say the expression system was used - but you probably also want to rearrange the sentences to more directly say what it was used for. Later, the word "the" is also missing.*

Thank you very much for your careful reading of the manuscript and valuable suggestions. We have revised this sentence to: "To further verify that Phe106 of CsAlaDC and Tyr111 of AtSerDC were key amino acid residues determining its substrate recognition in planta, we employed the *Nicotiana benthamiana* transient expression system. "

> *(19) Line 273: use "understand" instead of "elucidate" and instead of "we proposed a prediction test:" say "we designed a test of the prediction that..."*

Thank you very much for your careful reading of the manuscript. We have revised this sentence to: "In light of this observation, we postulated a hypothesis:"

> *(20) Line 301: I don't think "effectuate" is a word. Replace with something else.*

Thank you very much for your careful reading of the manuscript. We have revised the sentence as: " The biosynthetic pathway of theanine in tea plants comprises two consecutive enzymatic steps: alanine decarboxylase facilitates the decarboxylation of alanine to generate EA, while theanine synthetase catalyzes the condensation reaction between EA and Glu to synthesize theanine. "

> *(21) Line 307: replace "activity" with "ability"*

Thank you very much for your careful reading of the manuscript. We have corrected it in the revised manuscript.

> *(22) Line 322: I didn't find the discussion very useful. Much of it is simply a recap of the results - which is not necessary. The structural comparisons are overly descriptive without providing appropriate rationale or topic sentence structure so that the reader understands why certain details are emphasized. I think the manuscript would be much stronger if this section were not included or integreted more concisely into the results section where appropriate.*

Thank you for your constructive comments. We understand your concerns about the discussion section of our manuscript. We acknowledge that the discussion section has redundancies with the result. In response to this, we have revised this section to eliminate unnecessary repetition of the results.

> *(23) Line 369: "an amino acid devoid of the hydroxyl moiety present in Lys" - what does this mean? Lys does not have a hydroxyl functional group. Please correct so that the sentence makes sense.*

Thank you very much for your careful reading of the manuscript. This sentence states that the amino acid occupying the corresponding position in CsAlaDC is Phe, which lacks one hydroxyl functional group as compared to Lys. We have made modifications to the sentence as follows: "In contrast, the equivalent position in CsAlaDC is occupied by Phe, an amino acid lacking the hydroxyl group. This substitution enhances the hydrophobic nature of the substrate-binding pocket. "

> *(24) Line 370: "This structural nuance portends a predisposition for CsAlaDC to select the comparatively hydrophobic amino acid alanine as its suitable substrate." This sentence also makes no sense - please revise to use simpler language so the meaning is more clear.*

Thank you very much for your careful reading of the manuscript and valuable suggestions. We have revised the sentence as follows: " Consequently, CsAlaDC demonstrates a unique predilection, selectively binding Ala (an amino acid with comparatively hydrophobic properties) as its preferred substrate."

> *(25) Lines 376-384: This section makes several references to "catalytic rings." I have no idea what this term means? If the authors mean a loop structure in the enzyme - please use the term "loop"*

Thank you very much for your careful reading of the manuscript and valuable suggestions. We have corrected it in the revised manuscript.

> *(26) Line 396-397: The authors reference data that is not shown in the manuscript. Either show the data in the results section or do not mention.*

Thank you for your insightful comment regarding the unshown data referenced in the manuscript. We have included Supplementary figure 9 in the revised manuscript to display this data.

> *(27) Line 445-446: what is "mutation technology" - if the authors mean site-directed mutagenesis - please use the simpler and more recognizable terminology.*

Thank you very much for your careful reading of the manuscript and valuable suggestions. We have revised the sentence as follows: "Based on the findings of this study, site-directed mutagenesis can be employed to modify enzymes involved in theanine synthesis. This modification enhances the capacity of bacteria, yeast, model plants, and other organisms to synthesize theanine, thereby facilitating its application in industrial theanine production."

> ***Reviewer #3 (Public Review):***
>
> *In the manuscript titled "Structure and Evolution of Alanine/Serine Decarboxylases and the Engineering of Theanine Production," Wang et al. solved and compared the crystal structures of Alanine Decarboxylase (AlaDC) from Camellia sinensis and Serine Decarboxylase (SerDC) from Arabidopsis thaliana. Based on this structural information, the authors conducted both in vitro and in vivo functional studies to compare enzyme activities using site-directed mutagenesis and subsequent evolutionary analyses. This research has the potential to enhance our understanding of amino acid decarboxylase evolution and the biosynthetic pathway of the plant specialized metabolite theanine, as well as to further its potential applications in the tea industry.*

Thank you very much for taking the time to review this manuscript. We appreciate all your insightful comments.

> ***Reviewer #3 (Recommendations For The Authors):***
>
> *The additional material added by the authors addresses some of the previously raised questions and enhances the manuscript's quality. However, certain critical issues we*

> *pointed out earlier remain unaddressed. Some of the new data also raises new questions. To provide readers with more comprehensive data, the authors should include additional quantitative data and convert the data presented in the reviewer's comments into supplemental figure format.*

Thank you for acknowledging the improvements in the revised manuscript and providing further valuable feedback. We understand your concern about the critical issues that have not been fully addressed and the new questions raised by some of the newly added data. We have strived to address these issues with additional analysis and clarification in our subsequent revision. Regarding your suggestion for more quantitative data and converting the data mentioned in the reviewer's comments into a supplemental figure format, we agree that this would provide a more comprehensive view of the results. We have reformatted the relevant data into supplemental figures to enhance the clarity and accessibility of information. We are grateful for the time and effort you have dedicated to improving our manuscript.

> ** Page 5 & Figure 1B*
>
> *"As expected, CsSerDC was most closed to AtSerDC, which implies that they shared similar functions. However, CsAlaDC is relatively distant from CsSerDC."*
>
> *: In Figure 1B, CsSerDC and AtSerDC are in different clades, and this figure does not show that the two enzymes are closest. To provide another quantitative comparison, please provide a matrix table showing amino acid sequence similarities as a supplemental table.*
>
> *Comment: I don't believe that a 1.29% difference between 86.21% and 84.92% in amino acid similarity is statistically significant. Although the authors have rephrased the original sentence, it's improbable that this small 1.29% difference can explain the observed distinction.*

Many thanks. We have carefully considered your comments. Indeed, the 1.29% difference in amino acid similarity cannot reflect the functional difference between the AlaDC and SerDC proteins. We have deleted the relevant descriptions in the revised manuscript.

> ** Page 6, Figure 2, Page 23 (Methods)*
>
> *"The supernatants were purified with a Ni-Agarose resin column followed by size-exclusion chromatography."*
>
> *: What kind of SEC column did the authors use? Can the authors provide the SEC elution profile comparison results and size standard curve?*
>
> *Comment: The authors should include the SEC elution profiles as a supplemental figure or incorporate them as a panel in Figure 2. Furthermore, they should provide a description of the oligomeric state of each protein in this experiment. Additionally, there is a significant difference between CsSerDC (65.38 mL) and CsAlaDC (74.37 mL) elution volumes. Can this difference be explained structurally? In comparison to the standard curve of molecular weight provided by the authors, it appears that these proteins are at least homo-tetramers, which contradicts the description in the text. This should be re-evaluated and clarified.*

Thank you very much for your careful reading of the manuscript and valuable suggestions. We have included the SEC elution profile in Supplemental figure 1A and added descriptions of the oligomeric states of proteins in the revised manuscript. CsSerDC was eluted at 65.38 mL, corresponding to a molecular weight of 292 kDa, which is five times the monomeric protein (54.7 kDa). However, due to the absence of CsSerDC crystal structure, it remains uncertain whether the protein forms a pentamer. AtSerDC was eluted at 72.25 mL, with a corresponding molecular weight of 155 kDa, which is 3.3 times the monomer (47.3 kDa). CsAlaDC was eluted at 74.37 mL, with a corresponding molecular weight of 127 kDa, which is 2.7 times the monomer (47.3 kDa). The elution profiles suggest that AtSerDC and CsAlaDC potentially exist in homotrimeric form. This observation stands in contradiction to our subsequent findings where the protein manifests in a dimeric structure. A plausible explanation could be the non-ideal spherical shape of the protein. Under such circumstances, the hydrodynamic radius of the protein could supersede its actual size, potentially leading to an overestimation of the molecular weight on the size-exclusion chromatography [ref].

References:

Burgess, R. R. (2018) A brief practical review of size exclusion chromatography: Rules of thumb, limitations, and troubleshooting. *Protein Expression and Purification.* 150, 81-85.

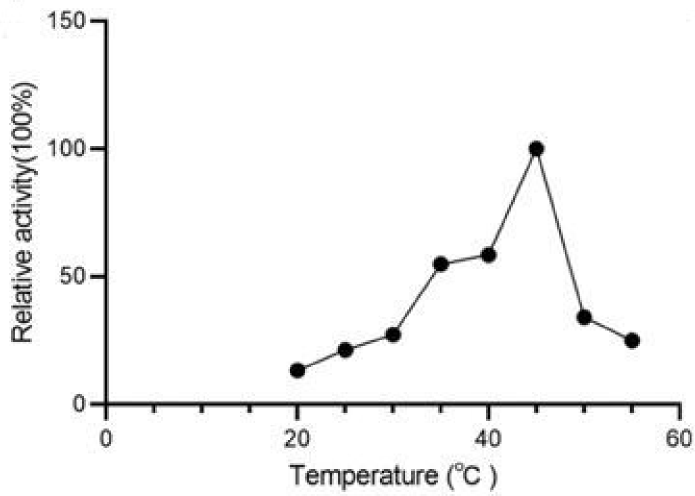
Erdner J. M., *et al*. (2006) Size-Exclusion Chromatography Using Deuterated Mobile Phases. *Journal of Chromatography A*. 1129(1):41–46.

> ** Page 6 & Page 24 (Methods)*
>
> *"The 100 µL reaction mixture, containing 20 mM substrate (Ala or Ser), 100 mM potassium phosphate, 0.1 mM PLP, and 0.025 mM purified enzyme, was prepared and incubated at standard conditions (45 {degree sign}C and pH 8.0 for CsAlaDC, 40 {degree sign}C and pH 8.0 for AtSerDC for 30 min)."*
>
> *(1) The enzymatic activities of CsAldDC and AtSerDC were measured at two different temperatures (45 and 40 {degree sign}C), but their activities were directly compared. Is there a reason for experimenting at different temperatures?*
>
> *(2) Enzyme activities were measured at temperatures above 40{degree sign}C, which is not a physiologically relevant temperature and may affect the stability or activity of the proteins. At the very least, the authors should provide temperature-dependent protein stability data (e.g., CD spectra analysis) or, if possible, temperature-dependent enzyme activities, to show that their experimental conditions are suitable for studying the activities of these enzymes.*
>
> *Comment: I appreciate the authors for including temperature-dependent enzyme activity data in their study. However, it remains puzzling that plant enzymes were tested at a physiologically irrelevant temperature of 40 and 45 degrees Celsius. Additionally, it may not be appropriate to directly compare enzyme activity measurements at different temperatures. Furthermore, the data at 45 degrees in panel A appears to be an outlier, which contrasts with the overall trend observed in the graph.*

We appreciate your point regarding the testing temperatures for plant enzymes. We fully appreciate the importance of conducting experiments under physiologically relevant conditions. But the intent behind operating at these elevated temperatures was to assess the thermal stability of the enzymes, which can be a valuable characteristic in certain applications, such as industrial production processes, and does not necessarily reflect their physiological conditions. Our findings indicate that CsAlaDC exhibits its peak activity at 45 °C. This result aligns with previously reported data in the literature [Bai, P. *et al*. (2021) figure 4e], thus bolstering our confidence in the reliability of our experimental outcomes.

Author response image 1.

Relative activity of CsAlaDC at different temperatures.



> ** Pages 6-7 & Table 1*
>
> *(1) Use the correct notation for Km and Vmax. Also, the authors show kinetic parameters and use multiple units (e.g., mmol/L or mM for Km).*
>
> *(2) When comparing the catalytic efficiency of enzymes, kcat/Km (or Vmax/Km) is generally used. The authors present a comparison of catalytic activity from results to conclusion. A clarification of what results are being compared is needed.*
>
> *Comment: The authors are still comparing catalytic efficiency solely based on the Vmax values. As previously suggested, it would be advisable to calculate kcat/Km and employ it for comparing catalytic efficiencies. Furthermore, based on the data provided by the authors, I conducted a rough calculation of these catalytic efficiencies and did not observe a significant difference, which contrasts with the authors' statement, "These findings indicated that the catalytic efficiency of CsAlaDC is considerably lower than that of both CsSerDC and AtSerDC." This discrepancy requires clarification.*

We want to express our sincere appreciation for your meticulous review and constructive suggestions. We understand the importance of accurately comparing catalytic efficiencies using Kcat/Km values, rather than solely relying on Vmax values. Following your suggestion, we recalculated Kcat/Km to reanalyze our results. The computed Kcat/Km for CsSerDC and AtSerDC are 152.7 s-1 M-1 and 184.6 s-1 M-1, respectively. For CsAlaDC, the calculated Kcat/Km is 55.7 s-1 M-1. Therefore, the catalytic efficiency of CsSerDC and AtSerDC is approximately three times that of CsAlaDC. What we intended to convey was that the Vmax of CsAlaDC is lower than that of CsSerDC and AtSerDC. Our description in the manuscript was not accurate, and we have addressed this in the revised version.

> ** Pages 9 & 10*
>
> *"This result suggested this Tyr is required for the catalytic activity of CsAlaDC and AtSerDC."*
>
> *: The author's results are interesting, but it is recommended to perform the experiments in a specific order. First, experiments should determine whether mutagenesis affects the protein's stability (e.g., CD, as discussed earlier), and second, whether mutagenesis*

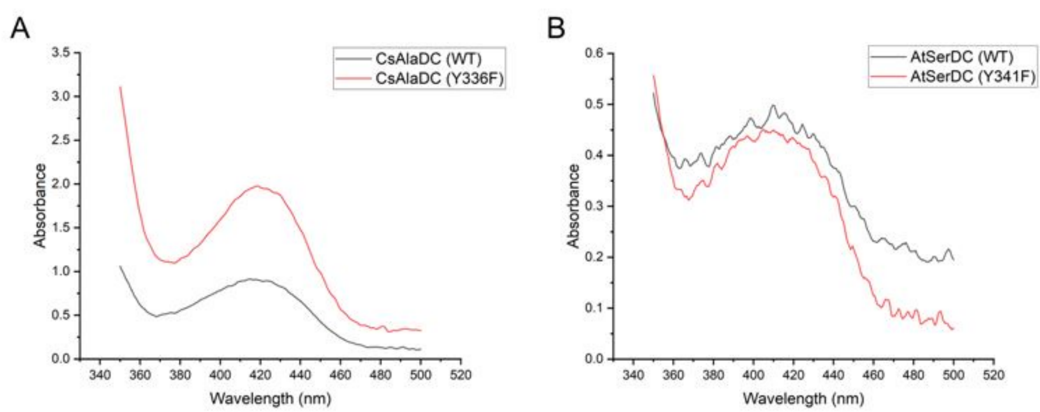
> *affects ligand binding (e.g., ITC, SPR, etc.), before describing how site-directed mutagenesis alters enzyme activity. In particular, the authors' hypothesis would be much more convincing if they could show that the ligand binding affinity is similar between WT and mutants.*
>
> *Comments: While it is appreciated that you have included CD and UV-vis absorption spectra data, it would be more beneficial to provide quantitative data to address the previously proposed binding affinity. I also recommend presenting the data mentioned in the reviewer's comments as a supplementary figure for better clarity and reference.*

Thank you for your valuable feedback and suggestions. I agree that providing quantitative data would lend more support to our findings and better address the proposed binding affinity.

It is generally acknowledged that proteins complexed with PLP exhibit a yellow hue, and the ligand PLP forms a Schiff base structure with the ε-amino group of a lysine residue in the protein, with maximum absorbance around 420 nm. However, during our protein purification process, we observed that the purified protein retained its yellow coloration, even when PLP wasn't introduced into the purification buffer. Subsequent absorbance measurements revealed that the protein exhibited absorbance within the aforementioned wavelength (420 nm) (the experimental results are shown in the following figures), implying an inherent presence of the PLP ligand within the protein. This could have resulted from binding with PLP during the protein's expression in *E. coli*. Consequently, due to this inseparability between the protein and the ligand, obtaining quantitative data through experimental means becomes unfeasible.

Author response image 2.

(A) Absorption Spectra of CsAlaDC (WT) and CsAlaDC (Y336F). (B) Absorption Spectra of AtSerDC (WT) and AtSerDC (Y341F).



Regarding your suggestion about presenting the data mentioned in the reviewer's comments as a supplementary figure, we agree that it is an excellent idea. We have prepared supplementary figure 7 and supplementary figure 8 accordingly, ensuring that they present the required data.

> ** Page 10*
>
> *"The results showed that 5 mM L-DTT reduced the relative activity of CsAlaDC and AtSerDC to 22.0% and 35.2%, respectively"*

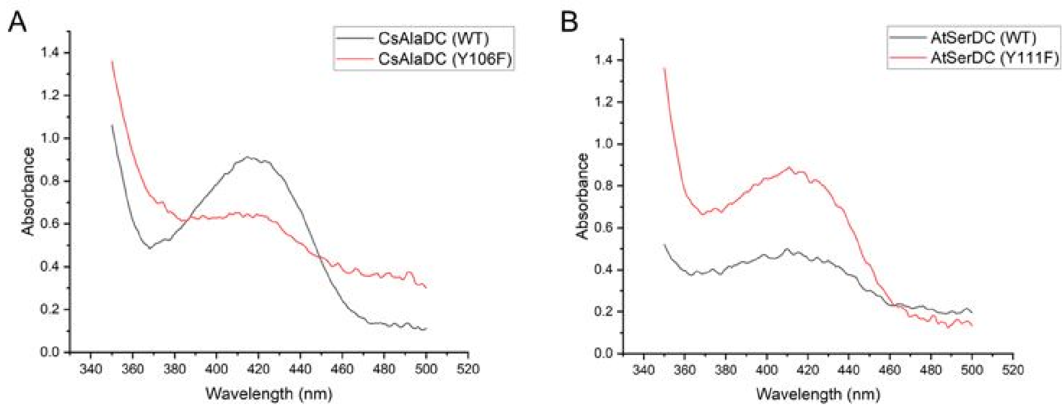
> *: The authors primarily use relative activity to compare WT and mutants. Can the authors specify the exact experiments, units, and experimental conditions? Is it Vmax or catalytic efficiency? If so, under what specific experimental conditions?*
>
> *Response: "However, due to the unknown mechanism of DTT inhibition on protein activity, we have removed this part of the content in the revised manuscript."*
>
> *Comment: I believe this requires a more comprehensive explanation rather than simply removing it from the text.*

Although we have observed that DTT is capable of inhibiting enzyme activity, at present, we are unable to offer a comprehensive explanation for the inhibitory effect of DTT on enzyme activity in terms of its structural and catalytic mechanisms. Further research is required to elucidate the mechanism of action of DTT. It is worth noting, however, that our study does not emphasize investigating the specific inhibitory mechanisms of DTT on enzyme activity. Furthermore, the existing findings do not provide an adequate explanation for the observed phenomenon, leading us to exclude this particular aspect from the content.

> ** Pages 10-12*
>
> *: The identification of 'Phe106 in CsAlaDC' and 'Tyr111 in AtSerDC,' along with the subsequent mutagenesis and enzymatic activity assays, is intriguing. However, the current manuscript lacks an explanation and discussion of the underlying reasons for these results. As previously mentioned, it would be helpful to gain insights and analysis from WT-ligand and mutant-ligand binding studies (e.g., ITC, SPR, etc.). Furthermore, the authors' analysis would be more convincing with accompanying structural analysis, such as steric hindrance analysis.*
>
> *Comment: While it is appreciated that you have included UV-vis absorption spectra data, it would be more beneficial to provide quantitative data to address the previously proposed binding affinity. I also recommend presenting the data mentioned in the reviewer's comments as a supplementary figure for better clarity and reference.*
>
> *Response: Thank you for your valuable feedback and suggestions. Given that the protein forms a complex with PLP during its expression in* E. coli *and cannot be dissociated from it, obtaining quantitative data via experimental protocols is rendered impracticable.*

Author response image 3.

(A) Absorption Spectra of CsAlaDC (WT) and CsAlaDC (F106Y). (B) Absorption Spectra of AtSerDC (WT) and AtSerDC (Y111F).



Mutant proteins and wild-type proteins exhibited absorption bands at 420 nm, suggesting the formation of a Schiff base between PLP and the active-site lysine residue.

Regarding your suggestion about presenting the data mentioned in the reviewer's comments as a supplementary figure, we have prepared supplementary figure 7 and supplementary figure 8 accordingly, ensuring that they present the required data.

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